Assessing Deep Learning Techniques for Ice Hockey Human Activity Recognition from Game Simulated Inertial Measurement Unit Data: A Comparative Analysis of CNN, Transformer, and Hybrid Models

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Abstract

The introduction of artificial intelligence (AI) has revolutionized the way sports data is collected and analyzed, opening new possibilities for what can be achieved in sports science research and performance. Human activity recognition (HAR) in ice hockey had traditionally relied on manual annotation and analysis from humans, a time-consuming task prone to error and bias. By partnering AI algorithms with movement data from wearable sensors, HAR models can be developed and implemented to provide deeper, more efficient, and more meaningful insights into hockey data. The objectives of this research were to compare three distinct deep learning neural networks in their abilities to classify eight ice hockey player events using wearable accelerometer data and to determine the optimal sensor configuration for this task. To accomplish these objectives, this study recorded 48 hockey players performing shots and other hockey events in a game-simulated environment using 17 wearable inertial measurement units (IMU). Video footage of the participant trials was used as ground truth for labelling the IMU data. The segmented and labelled acceleration data were then used to train and test three deep-learning model architectures: a traditional approach (Convolutional Neural Network (CNN)), a modern approach (Transformer Network), and a hybrid approach combining the benefits of both models (CNN-Transformer Hybrid). To further the investigation, the three model architectures were trained and tested on three different sensor configurations. These configurations included a 17-sensor configuration featuring IMUs on all body segments, a 3-sensor configuration featuring IMUs on the hands and sternum, and a 2-sensor configuration featuring an IMU on each hand. Results revealed that the Hybrid models outperformed the other models, achieving the highest mean F1 score across all three sensor configurations (81.9%), followed by the CNN models (79.5%) and the Transformer models (74.4%). The highest-performing individual model was the

3-sensor Hybrid model, reaching an F1 score of (83.7%). The models all demonstrated their ability to differentiate between tasks with different movements (e.g. *slap shot* vs *pass*) but struggled to differentiate between tasks with similar movement patterns (e.g., *slap shot* vs *one-timer*). Findings from this study can have a positive impact on the advancement of how movement data is detected and analyzed in real-game environments. These advancements have important implications for coaches, players, and analysts by supporting the development of AI technology for injury detection and rehabilitation, real-time performance tracking, and technique optimization to enhance player performance.

Abrégé

L'introduction de l'intelligence artificielle (IA) a révolutionné la manière dont les données sportives sont collectées et analysées, ouvrant de nouvelles possibilités en matière de recherche et de performance dans le domaine des sciences du sport. La reconnaissance de l'activité humaine (RAH) dans le hockey sur glace reposait traditionnellement sur l'annotation et l'analyse manuelle par des humains, une tâche fastidieuse qui est sujette à des erreurs et à des biais. En associant des algorithmes d'IA à des données de mouvement provenant de capteurs portables, il est possible de développer et de mettre en œuvre des modèles de reconnaissance de l'activité humaine afin d'obtenir des informations plus approfondies, efficaces et plus significatives sur les données relatives au hockey. Les objectifs de cette recherche étaient de comparer trois réseaux neuronaux d'apprentissage profond distincts dans leurs capacités à classer huit événements de joueurs de hockey sur glace à l'aide de données d'accéléromètre portables et de déterminer la configuration de capteur optimale pour cette tâche. Pour atteindre ces objectifs, cette étude a enregistré 48 joueurs de hockey effectuant des tirs et d'autres événements de hockey dans un environnement de simulation de jeu à l'aide de 17 unités de mesure inertielle (UMI) portables. Les séquences vidéo des essais des participants ont été utilisées comme vérité de terrain pour l'étiquetage des données UMI. Les données d'accélération segmentées et étiquetées ont ensuite été utilisées pour développer et comparer trois architectures de modèles d'apprentissage profond: une approche traditionnelle (réseau neuronal convolutif (CNN)), une approche moderne (réseau Transformer) et une approche hybride combinant les avantages des deux modèles (CNN-Transformer Hybride). Pour approfondir l'étude, les trois architectures de modèles ont été entraînées et testées sur trois configurations de capteurs différentes. Ces configurations comprenaient une configuration à 17 capteurs comprenant des UMI sur tous les segments du

corps, une configuration à 3 capteurs comprenant des UMI sur les mains et le sternum, et une configuration à 2 capteurs comprenant une UMI sur chaque main. Les résultats ont révélé que les modèles hybrides étaient plus performants que les autres modèles, obtenant le score F1 moyen le plus élevé pour les trois configurations de capteurs (81,9 %), suivis par les modèles CNN (79,5 %) et les modèles Transformer (74,4 %). Le modèle individuel le plus performant était le modèle hybride à trois capteurs, qui a obtenu un score F1 de (83,7 %). Les modèles ont tous démontré leur capacité à différencier des tâches avec des mouvements différents (par exemple, *tir frappé* versus *passe*) mais ont eu de la difficulté à différencier des tâches avec des modèles de mouvements similaires (par exemple, *tir frappé* versus *one-timer*). Les résultats de cette étude peuvent avoir un impact positif sur l'évolution de la détection et de l'analyse des données de mouvement dans les environnements de jeu réels. Ces progrès ont des implications importantes pour les entraîneurs, les joueurs et les analystes en soutenant le développement de la technologie de l'IA pour la détection des blessures et la rééducation, le suivi des performances en temps réel et l'optimisation des techniques afin d'améliorer les performances des joueurs.

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Philippe J. Renaud, Department of Kinesiology and Physical Education, McGill University, acted as the lab research assistant, contributing to the research design and collection of data. Philippe C. Dixon and Julie Côté, Department of Kinesiology and Physical Education, McGill University, were members of the thesis committee.

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List of Abbreviations

HAR	Human Activity Recognition
AI	Artificial Intelligence
CNN	Convolutional Neural Network
LSTM	Long-Short Term Memory
RNN	Recurrent Neural Network
IMU	Inertial Measurement Unit
ASC	All Sensor Configuration
SHSC	Sternum and Hands Sensor Configuration
HSC	Hands Sensor Configuration
SS	Slap Shot
WS	Wrist Shot
BH	Back Hand
OT	One-timer
P	Pass
PC	Puck Control
NP	No Puck

1. Introduction

The introduction of artificial intelligence (AI) has revolutionized the way sports data is collected and analyzed, opening new possibilities for what can be achieved in the field of sports science. With the competitive sporting industry continuing to become more reliant on technology and analytics, activity recognition systems can provide players, coaches, managers, and stakeholders with valuable information that otherwise would not be accessible. In the fast-paced, dynamic sport of ice hockey, traditional data collection and analysis methods require manual human interaction, leading to potentially inefficient, inaccurate, and biased data processing. Modern AI models can automate these processes, allowing for quicker analyses and can provide deeper, more meaningful insights into the data.

Human Activity Recognition (HAR) involves identifying and classifying different types of human movements or activities based on data collected from movement sensors or videos. It is particularly important in sports science because it allows for the analysis of specific athletic movements, aiding in performance evaluation, technique refinement, injury prevention, and rehabilitation. HAR is also widely applicable in other domains, such as healthcare, rehabilitation, and everyday activity monitoring (Al-qaness et al., 2024; Gupta, 2021; Schrader et al., 2020) . To successfully develop HAR models, accurate and detailed data of the specific movement of interest must be recorded.

A longstanding challenge in sports biomechanics has been the accurate collection of realistic movement data without interfering with an athlete's performance. Advancements in wearable technology have allowed sports scientists to overcome this challenge by applying small, lightweight wearable sensors to the athletes to record their movements. These sensors, called Inertial Measurement Units (IMUs), can provide rich, detailed kinematic data about an

athlete's performance in their natural sporting environment but require further analysis to extract meaningful information. By leveraging AI techniques to make sense of this complex, detailed, and high-dimensional data, HAR models can capture detailed temporal patterns in movements and provide insights that can be otherwise challenging to obtain.

The sport of ice hockey poses a unique challenge to HAR due to its continuous dynamic gameplay, enclosed low-friction playing surface, and protective equipment worn by players. These challenges limit the feasibility of collecting ice hockey kinematic data using traditional fixed camera-based motion capture systems. Fortunately, wearable wireless IMUs have proven to be a suitable alternative for collecting detailed multi-segment body data in ice hockey applications (Hardegger et al., 2015; Khandan et al., 2022; Vleugels et al., 2021).

Many studies have employed machine and deep learning models for HAR with high levels of success (Cust et al., 2019), with one of the most used models being the Convolutional Neural Network (CNN). While CNNs have demonstrated their effectiveness, the field of HAR is continually evolving, with new methods frequently being introduced and adopted. One of the most promising recent advancements is the Transformer network. First introduced by Google in 2017 for natural language translation tasks, Transformer networks leverage a self-attention mechanism that enables them to capture long-range dependencies, improving how complex relationships in data are modelled. Initially developed for text data, Transformers have demonstrated great promise in time-series analysis and are now being applied to HAR tasks (Dirgová Luptáková et al., 2022). Despite their growing popularity, their use in ice hockey biomechanics remains largely unexplored.

Therefore, this study aimed to develop and compare the performance of a traditional CNN model, a Transformer model, and a Hybrid model combining both approaches, in

classifying eight different ice hockey events. This was accomplished using data collected from three different IMU configurations in a game-simulated environment from 48 ice hockey players.

Findings from this study could have significant implications for sports science and biomechanics. They could enhance our understanding of how AI and wearable technology can be used to record and analyze complex athletic movements in dynamic sports like ice hockey. Additionally, this research could contribute to the development of more advanced performance analysis tools, enabling athletes and coaches to gain actionable insights to optimize analytics, training, and rehabilitation methods. By examining the use of Transformer and Hybrid networks in a new context, this study aims to advance the application of AI in sports analysis.

2. Review of Literature

2.1 Introduction to Human Activity Recognition

The field of Human-Computer Interaction is rapidly evolving as technology becomes increasingly central to daily life. In biomechanics, HAR is a common Human-Computer Interaction tool that assists in recording and analyzing human movement. HAR is the automatic detection of human movements using data from motion-capture technologies and AI techniques. Many HAR models follow the Activity Recognition Chain framework, a sequence that includes data acquisition, pre-processing, data segmentation, feature extraction, model training, event classification, decision fusions, and performance evaluation (Bulling et al., 2014). HAR methods are developed to detect and assess various human actions, such as walking, running, or more complex tasks, in an efficient and objective manner. Before these technologies were introduced, humans had to manually complete tasks like identifying and annotating movements either in real-time or from video footage. They would then provide their own subjective assessment of the performance of the movement, which can be limited and influenced by the knowledge of the individual. These processes can be very time-consuming and are prone to human error and bias.

HAR has demonstrated practical value in a breadth of applications. Some of these include the assessment of upper limb function in breast cancer patients before and after surgery using wearable inertial sensors (Emmerzaal et al., 2024), monitoring the rehabilitation of daily tasks for elderly individuals (Schrader et al., 2020), as well as detecting usual and unusual activities in security surveillance applications using video data (Sunil et al., 2021). These examples showcase its versatility in identifying simple tasks across various fields. However, HAR is also being extended to settings involving complex movements, particularly in sports.

The increasing integration and dependence on technology in sports can help players and trainers monitor activity levels, track performance results, and evaluate form, all of which can enhance their training efficiency and athletic performance. These techniques can also assist in highlighting factors that can lead to certain sports injuries (Chen & Yuan, 2021) and monitor players' rehabilitation from injury (Zhan, 2024). The challenge in sports movement recognition often lies in the intensity and complexity of the tasks exhibited in sports. Many sports movements involve multiple overlapping body segments and elaborate, intense sequences. Individuals' task execution can vary between people based on their anatomy, biomechanics, injury history, and skill level, meaning that two people might achieve the same result by performing an action differently. HAR algorithms must be able to detect both large dynamic movements as well as subtle differences in varying sports task execution (Mekruksavanich et al., 2024). Advancements in technology have enabled sensors to be made smaller, more portable, and more affordable, making them the primary recording technology for HAR research (Gupta, 2021). With the dynamic movement patterns and unpredictability of competitive sports, these improvements in wearable inertial sensors have allowed for the recording of meaningful information about the performance of various sporting tasks in the natural field of play (Cust et al., 2019).

2.2 Human Activity Recognition in Recognition in Sport

Technical and tactical sports analysis has evolved over the years. Traditional methods involved coaches or trainers manually recording notes and statistics during gameplay. These methods are prone to subjective biases and errors and are limited by the abilities of the annotators. The automatic detection of sport-specific movements overcomes the limitations

associated with traditional methods and supports the accurate, efficient, and objective assessment of an athlete's performance (Cust et al., 2019). HAR in sports can provide valuable insight into athletic performance, training optimization, and injury prevention, playing a pivotal role in modern sports science. Modern technology enables coaches to monitor their players' performance using sensors that can track metrics like heart rate, speed, movement patterns in training, and positioning in games. For example, HAR might reveal that a hockey player performs 80% of their slap shots from a certain location on the ice or takes a longer time than average to pivot from forward skating to backward skating. These sorts of insights could allow coaches to optimize an athlete's training plan and objectively monitor and track their improvements.

Various technologies have been implemented in the field of HAR in sports, typically using movement data from either videos or wearable sensors. Early sports event detection typically involved vision-based systems and relied on manual viewing and annotating video footage to track events and actions. Modern video-based approaches employ different computer vision techniques and algorithms that leverage spatial and temporal features from RGB pixel data to recognize movements. Liu (2021) developed a 3D Convolutional Neural Network (CNN) to classify basketball movements using pixel video footage with 80% recognition accuracy (Liu, 2021). Vats and colleagues (2019) used an Optical Flow algorithm for ice hockey action recognition, which measured pixel-wise displacement over time to calculate the direction and velocity of movement for each pixel, providing insights into the movement of body parts as well as understanding motion and trajectories. Their proposed model achieved 90.48% test accuracy on a hockey action recognition dataset (Vats et al., 2019). Video-based approaches, however, are

limited to various factors, including lighting, field of view, complexity of movement, and capture area, which often limits the ability to capture data in the actual field of play (Cust et al., 2019).

Alternate approaches that fasten sensors to an athlete's body can help mitigate these limitations. Inertial measurement units (IMU) are tri-axial sensors that are fitted with an accelerometer, gyroscope, and magnetometer, recording acceleration, angular velocity, and positional data in all three axes. These sensors can be placed on a desired position on an athlete's body to record 3-dimensional motion in real-time. Wireless IMU systems can be worn by athletes or integrated into their equipment, allowing data capture in real-time within their natural sporting environment. Recent advancements in wearable technology have increased their popularity, with sensors being embedded in various devices ranging from smartphones and wearable accessories to smart garments. IMUs from smartphones have been used to classify various alpine skiing activities (Azadi et al., 2022), and data from watch IMUs has been used in basketball and badminton movement classification (Zhuang & Xue, 2019). Similarly, Smart Textile Systems, garments with embedded sensors, have been used in various sporting, entertainment, and medical applications (Xu et al., 2022), such as vests with IMUs and EMGs to monitor postural control (Martins et al., 2024). These diverse recording methods have enabled a more adaptive approach to capturing and analyzing movement data across various sports and settings, advancing the capabilities and accuracy of sports event recognition, leading to enhanced insight into gameplay analytics and performance details.

2.3 Deep Learning for Event Recognition

Modern activity recognition requires AI algorithms to learn and detect desired actions from raw movement data. The challenge with HAR in sports is that many sporting actions are

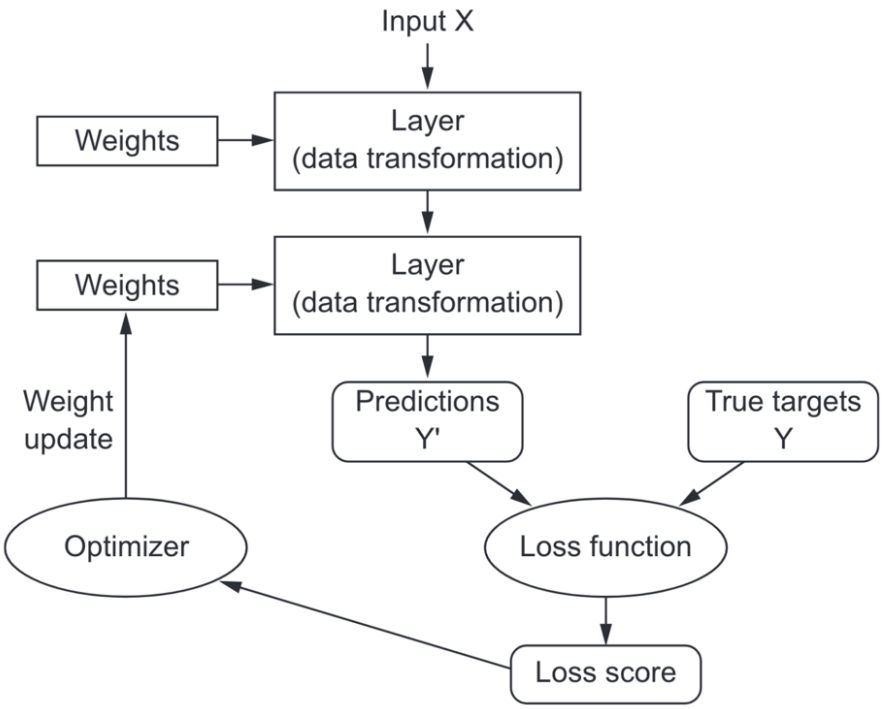
complex and dynamic in nature. Some movements are multi-segment, explosive actions, while others are small, subtle adjustments that impact the outcome of an event. This broad range of movement types demands highly efficient AI algorithms capable of automatically extracting features from sensor data without manual intervention, an area where traditional machine-learning techniques can fall short (Mekruksavanich et al., 2024). Machine learning models often rely on manual feature engineering, which requires human involvement and limits their efficiency and adaptability (Chollet, 2018; Gupta, 2021). Deep learning models, on the other hand, perform automatic feature extraction, simplifying the end-to-end model development process. What sets deep learning models apart is their ability to learn successive layers of increasingly meaningful representations. These models are well equipped to handle complex, high-dimensional temporal data like time series data from human activities that include aspects such as motion, transitions, sequencing, duration, intensity, and coordination (Mekruksavanich et al., 2024) and can more effectively learn from vast amounts of data (Alzubaidi et al., 2021; Gupta, 2021).

Supervised deep learning classification algorithms have become a fundamental part of the activity recognition chain in HAR in many sporting contexts. The goal of supervised learning is to map input data (e.g. a picture of a fruit) to a known target class (e.g. apple) using many samples of ground truth annotated data (e.g. fruit pictures with corresponding labels). Once a supervised learning model is trained, it can classify new, unseen data into one of the pre-determined classes. This is accomplished by forward-passing the input data through successive layers of data transformation (forward pass) (Chollet, 2018). In complex models, these layers contain millions of neurons with adjustable parameters (weights and biases) that are optimized in the training process to produce accurate final predictions. Once the input data is processed

through the multiple layers, a final output layer predicts the appropriate class for that input data. A loss function then generates a loss score based on how far off the predicted class is from the actual class, indicating how well the network performed on that example. Through a mechanism called backpropagation, the loss score is used as a feedback signal to calculate gradients, which inform the optimizer of the magnitude and direction needed to adjust parameters (weights and biases) in each layer. The learning rate determines how much the parameters are adjusted in response to the gradient, controlling the speed and stability of the learning process. These steps combine to reduce the loss in subsequent iterations. This workflow, called the training loop (Figure 1), repeats to continually minimize the loss score until it reaches convergence, where further updates to the parameters no longer significantly improve the model. Once the predicted output classes closely match the target classes, the loss function stabilizes near a minimum value, indicating that the model has learned underlying patterns in the data and the network has been trained (Chollet, 2018).

Figure 1

A Conceptual Neural Network Training Flow (Chollet 2018)



Note. Figure use permission obtained (April 11, 2025).

Most activity recognition tasks use multivariate time-series data with high spatial and temporal resolution, typically collected from inertial sensors or videos (Hammerla et al., 2016). This introduces the Time Series Classification challenge, which involves classifying sequences of data points based on how they behave over time (Gupta, 2021). For example, a time series classification model could analyze accelerometer sequences of hockey shots to classify them as either wrist or slap shots based on their unique movement patterns over time. This type of data influences the choice of suitable deep learning models for time series classification tasks, as they must be able to effectively model both temporal and spatial relationships in the data.

Convolutional Neural Networks (CNN) are among the most widely used models because of their ability to automatically extract spatial features, detect local patterns, and learn

hierarchical features in data (Gupta, 2021). While originally designed for 2-dimensional data and commonly utilized in fields such as computer vision, facial recognition, and speech processing (Alzubaidi et al., 2021), CNNs can be applied to 1-dimensional time series classification tasks. Using a modified architecture, a 1D CNN kernel moves along a single dimension of acceleration or gyroscope signal from a sensor to automatically extract spatial or local information. The architecture of a CNN typically consists of multiple blocks. These blocks are comprised of a convolutional layer which generates the feature map from the signal, a pooling layer which sub-samples the features, an activation function which maps the inputs to the outputs, a fully connected layer, and a loss function which reveals the distance of the prediction to the actual class (Alzubaidi, et al., 2021).

CNNs have proven to be highly effective for HAR, often outperforming traditional machine learning methods (Cust et al., 2019). Shakya and colleagues compared machine learning to deep learning techniques in their HAR capabilities (Sarbagya Ratna Shakya, Chaoyang Zhang, and Zhaoxian Zhou, 2018). They used two publicly available datasets containing daily actions, including walking, sitting, standing, and going upstairs, collected by five accelerometers placed on five different parts of the body. After comparing different models, results showed that the highest-achieving deep learning model, a CNN (99.16% accuracy), outperformed the highest-achieving machine learning model, K-nearest neighbours (97.65 % accuracy) (Shakya, et al., 2018).

Another commonly used model for HAR is Long Short-Term Memory (LSTM). These models are a form of Recurrent Neural Network (RNN) that are designed to learn and remember patterns in sequential data by storing information about previous time steps in a hidden state, making them useful for time-series models. However, in the backpropagation stage of RNNs, the

gradients are calculated by recursively multiplying the same weight matrix. As a result, the gradient can exponentially grow and explode when the multiplication involves large numbers, or it can exponentially shrink and vanish when the multiplication involves small numbers. Without proper gradient control, the model will not be able to learn (Pascanu et al., 2013). To mitigate this issue, LTSMs incorporate cell states for improved memory, as well as forget, input, and output gates for the flow of information to maintain relevant context in the network (LeCun et al., 2015). Combining CNNs and LSTMs in hybrid architectures is an increasingly popular approach, as it combines the spatial feature extraction of CNNs with the temporal sequence modelling capabilities of LSTMs, enhancing performance in complex HAR tasks (Deep & Zheng, 2019).

While CNNs and LSTMs are widely used models, they both possess limitations. Convolutional layers and pooling operations help CNNs excel at capturing local features, but they are prone to challenges when modelling long-range dependencies, which can exist in sequential time series data. LSTMs process sequences step-by-step, leading to longer training times and difficulty remembering information over extended periods. To address these issues, researchers created models using an attention mechanism, which helps the model decide which parts of a sequence are most important and focuses on them. This makes it easier for the model to understand and use the correct information for the specific task.

In the seminal paper titled “Attention is All You Need”, researchers at Google introduced the first Transformer model, which featured a self-attention mechanism at its core for sequential modelling (Vaswani et al., 2023). This self-attention mechanism replaces the sequential processing of recurrent networks and the localized focus of convolutional networks by parallelizing the computations of each element in a sequence, allowing every element to directly

attend to every other element within it. In the attention mechanism, sequences are first segmented into tokens, which are then transformed into query, key, and value representations. The queries and keys are compared to generate attention scores, which determine the importance of each token, influencing how much each value contributes to the model's output (Vaswani et al., 2023). Similar to how a word in a sentence is given significance based on the context of other words in that sentence, each element in a segment of time-series data bears significance based on the preceding and subsequent data points.

While originally designed for language translation tasks, innovation in Transformer architecture has led to significant gains in tasks ranging from natural language generation (Topal et al., 2023.) to complex biological challenges like protein sequence predictions (Chandra et al., 2023) to large-scale HAR tasks (Dirgová Luptáková et al., 2022). Dirgová Luptáková (2022) adapted a transformer model to classify accelerometer and gyroscope data from a dataset of 18 different activities, some of which combined consecutive tasks. The proposed model achieved 99.2% prediction success, proving that the attention mechanism was capable of detecting and promoting important features from the long, complex accelerometer signals, positively affecting the classifications of the many activities.

Since transformers were originally introduced for sequential tasks, their architecture typically follows an encoder-decoder format. In these architectures, the first section, called the encoder, processes an input sequence to generate a vector representation. The second section, called the decoder, uses the representation to generate an output sequence. For classification tasks, as shown in Dirgová Luptáková et al. (2022), only the encoder is needed because the classification can be performed directly on the hidden representations produced by the encoder, eliminating the need for a separate decoding step typically required for sequence generation.

2.4 Inertial Measurement Units

In recent years, IMUs have emerged as a powerful tool for recording and analyzing human movement data, particularly in sports science settings. These small, lightweight, wearable sensors can be secured to a specific area of interest on an athlete's body to record movement data while causing little to no interference in their gameplay. IMUs are self-contained units composed of accelerometers, gyroscopes, and magnetometers, quantifying acceleration, angular velocity, and direction and orientation, respectively, along one to three axes (Cust et al., 2019). Traditional optoelectrical motion capture systems restrict data collection to a confined recording space and require a clear line of sight to the reflective markers placed on the participants. Data collection for hockey research using optoelectrical systems typically requires researchers to set up in a lab space on artificial ice (Michaud-Paquette et al., 2011) or on a small section on real ice (Mazurek et al., 2024). These limitations impact the generalizability of ice hockey studies using camera-based motion capture systems, as wearing reflective markers means that participants cannot wear their full equipment, and they are unable to behave as they would in a real game due to the limited capture area. IMUs provide an effective solution to the limitations of optoelectrical motion capture systems while still capturing accurate kinematic data. The wireless, non-intrusive sensors are small enough to fit underneath equipment and allow participants to skate freely around the entire rink.

IMUs have improved the feasibility of recording and analyzing kinematic data in various sports settings, which are otherwise challenging with existing motion capture systems. Slopecki et al. (2023) used an IMU placed on the sacrum of para-swimmers to record and accurately estimate various swim stroke metrics. After comparing the IMU results to video-derived metrics, Slopecki concluded that a single IMU can effectively monitor the performance of a majority of

swim strokes for elite para-swimmers and that IMUs address the drawbacks of traditional two-dimensional video analysis that include water turbulence, parallax errors, and time-consuming data processing (Slopecki et al., 2023).

A study in 2020 validated the use of an IMU system for movement tracking in ice hockey. Denroche (2020) compared the joint kinematics of wrist and slap shots recorded from a 17-sensor Xsens MVN Link IMU system to a "gold standard" Vicon optoelectric motion capture system. Both systems were comparable about the X (sagittal) and Y (frontal) axes. Some differences were observed about the Z (transverse) axis, and there was disagreement with shoulder axial rotation, but these differences were attributed to the systems' separate calibration procedures and biomechanical models. Denroche further highlighted that the wearable Xsens IMU system enhances the feasibility and practicality of full Xsens-based hockey studies on real ice under game-like conditions (Denroche, 2020).

2.5 AI Using IMU Data

Raw IMU data is often very detailed and complex, requiring advanced processing techniques to extract meaningful insights. Recent studies have investigated the integration of IMUs with machine and deep learning models for a variety of classification and performance analysis tasks. A study on water polo demonstrated the ability of IMUs and AI to detect shots and passes using data collected from an IMU on the dominant wrist and sacrum of eight water polo players during six practices (Croteau et al., 2022). Two different models were trained and tested using the tagged IMU shot and pass data. Ground truth data was obtained by a researcher who video-recorded the data collection sessions and manually tagged the shots and passes. A support vector machine (SVM) achieved 94.4% sensitivity and 93.6% specificity, and an

artificial neural network achieved 93% sensitivity and 93.4% specificity (Croteau et al., 2022). Similarly, in the context of golf, researchers fastened accelerometers and gyroscopes to golf clubs to record the swing shapes of four professional and amateur golfers. Researchers trained a Deep CNN model to classify swing shapes, achieving a 95% F1 score and outperforming an SVM model that achieved an 87% F1 score (Jiao et al., 2018). F1 score is a metric for model evaluation, comprised of a weighted average of precision and recall scores. Further demonstrating the versatility of IMUs and AI in sports, soccer researchers placed a custom IMU in the cavity of soccer shoes and collected data in both a controlled exercise setting and in a live match. Four statistical features were extracted for each sensor axis (mean, variance, skewness, and kurtosis) and were used to train and test three different machine learning classifiers (SVM, Classification and Regression Tree, and Naïve Bayes). The highest-performing model was the SVM, achieving an overall mean classification rate of 84.2% (Schuldhaus et al., 2015).

The integration of IMUs and AI has shown success in sports possessing qualities similar to ice hockey, including the unique playing surface exhibited in water polo, the swing mechanics in golf, and the dynamic, continuous gameplay of soccer. Based on these similarities, ice hockey offers a practical application to leverage these technologies for advanced analytic or predictive tasks. This was demonstrated in a recent ice hockey study that introduced a method for instantaneous hockey stick fitting using IMU sensors and machine learning algorithms (Léger et al., 2022). Researchers embedded IMUs in the gloves of ten hockey players and recorded kinematic data of 80 shots from four sticks of different constructions. Principal components of the linear acceleration and angular velocity of both hands were used to train and test a K-nearest neighbours machine learning model, to classify players' shooting performance with different stick flexes, blade patterns, and kick points, as well as to determine which IMU channel was best

for each shot type and stick configuration. The model achieved 90-98% accuracy for classifying stick flex, blade pattern, and kick point for slap shots and 93-97% accuracy for wrist shots, demonstrating the applicability of IMUs and machine learning models for ice hockey stick fitting and shooting analysis (Léger et al., 2022).

2.6 Ice Hockey Event Classification

The sport of ice hockey is a high-speed, continuous game that requires its players to perform an array of actions, ranging from dynamic and explosive bursts to smooth, controlled, precise movements, while maintaining efficient transitions between actions. Many hockey skills are categorized as 'open' because they are not uniformly executed. Instead, they are adapted to meet the physical and environmental demands of the player's current position on the ice, requiring quick reactions to the constantly changing positions of opponents and teammates. Based on these demands, to effectively execute a hockey skill, players must be able to efficiently choose an action response to their changing environment based on a multitude of internal and external cues. These skills require coordination of timing, anticipation, balance, versatility, direction, accuracy, rhythm, reaction time, and speed (Turcotte, 2000).

Effectively analyzing hockey skills is essential for key sporting objectives like technique improvement, injury rehabilitation, or player evaluation; however, the nature of hockey makes this a challenge. The low-friction ice surface that the game is played on poses a unique set of challenges to both the performance and the performance monitoring of ice hockey tasks. Skating analysis is a unique challenge because sufficient ground reaction forces cannot be generated at the contact between the foot and the ground to propel the body forward, meaning conventional human walking and running patterns cannot be applied to skating (Pearsall, 2013). Further, setting up a motion capture system on ice in an arena’s cold, humid environment is a labour-

intensive and delicate process (Renaud et al., 2017). The full-body protective equipment worn by hockey players introduces an additional challenge, as it can compromise the accuracy of both marker-based and markerless motion capture systems. Furthermore, the speed and frequency of in-game movements make it challenging to accurately record and analyze actions through visual means alone.

To address these challenges, modern technology, such as IMUs, can be employed to assist in recording and analyzing the numerous complex movements inherent to hockey. Inertial sensors have proven to be sufficient to record even the subtle kinematic differences that exist between males and females that lead to different performance outcomes (Holman, 2024). That study used 17 IMUs to record the kinematic differences between males and females in tight turns, revealing that males exhibit a greater internal rotation of the outside hip and internal rotation of the inside ankle, which may optimize cutting angles (Holman, 2024).

Combining this sensor technology with artificial intelligence techniques can provide further detailed and specific insights into hockey task execution, overcoming the limitations of traditional methods. These techniques are demonstrated by Hardegger et al. (2015), who used IMUs placed on participants' sternums, wrists, and sticks to capture movement data from 19 hockey players and implemented a feature-based classification model to identify different gameplay events. The accelerometer and gyroscope data were used to train a Random Forest classifier with varying window sizes. While the model could effectively classify shots and hits, it struggled to identify the direction of turns, which subsequently affected the classifiers' performance in identifying other events. Additionally, the classifier was unable to detect longer-term events, such as whether or not the player was in possession of the puck. The model

achieved an F1 score of 70% when detecting events in the determined optimal window size of 0.6s, and a score of 86% in determining if the player was on the ice or not (Hardegger, 2015).

More recently, Tremblay (2022) developed a CNN to classify different hockey shots based on acceleration data from IMUs worn by elite players. In this study, 43 elite hockey players were fitted with a 17-sensor IMU system recording at 240 Hz on real ice. Each participant performed wrist shots (n=15), slap shots (n=15), one-timers (n=10), backhand shots (n=10), passes (n=5), stick handling trials (n=5) and a rest on the bench (n=1). For all non-one-timer trials, players were instructed to skate from the middle of the ice toward the net and perform their shot in a designated shooting zone. For one-time trials, players stood stationary above the faceoff dot on the side corresponding to their handedness to execute their one-timers. This study compared two models: a full-body 17-sensor configuration and a hands-only two-sensor configuration. The CNN trained on the data from the 17-sensor configuration achieved an F1 score of 97.1%, outperforming the 93.6% F1 score from the two-hand sensor configuration. (Tremblay, 2022). These findings demonstrate the viability of wearable sensor technology and AI models to detect and classify dynamic hockey movements, enabling future innovation in hockey science and development.

2.7 Gaps in the Literature

A key limitation of the classification study by Tremblay (2022) is that the data were collected from players performing shots in a controlled environment where each shot was executed in a similar manner. As a result, participants exhibited a consistent movement pattern for their shots, reducing the variability and unpredictability of their trials. This meant that the model learned a very limited scope of shot techniques. The test also did not account for other

natural gameplay events (i.e., stick handling, skating without the puck, passing between shots, puck battles, and resting on the bench). While this study demonstrated the feasibility of developing a hockey event classification model, its generalizability to real-game environments could be limited. In a live hockey game, players adapt their shooting technique based on various environmental factors, including the time they have with the puck, the puck's path to the net, their position on the ice, and the goalie's positioning in the net. These various in-game factors will change a player's shot kinematics. To create a model that could successfully be used in a real game setting, it needs to be trained and evaluated using data that simulates the shooting and playing environment of a real game.

In ice hockey, many movements appear to be performed with a similar motion, but minor biomechanical adjustments can greatly alter the performance outcome. The CNN approach that Tremblay (2022) proposed demonstrated great success for shots in a controlled setting, as those networks excel at detecting local patterns. However, CNNs have a fixed receptive field that can limit their ability to detect the broader contextually relevant cues of longer-range dependencies in hockey movements. As a result, CNNs may struggle to distinguish the differences in the subtle yet crucial variations in movements that exist in the game of hockey. The Transformer's attention mechanism is able to process an entire sequence at once, enabling it to capture a broader range of patterns in the data. This is crucial for ice hockey movement data, where the outcome of an action often depends on what occurs before and after it. Both CNNs and Transformer models possess strengths in feature extraction and contextual models. For tasks that require both local feature recognition and long-range dependency analysis, leveraging the strengths of both approaches in a hybrid model could yield enhanced classification performance.

Building on the previous research in this field will continue to transform how ice hockey data is recorded, detected, and analyzed. Addressing these research gaps by adopting modern AI techniques with game-realistic data can further optimize the accuracy of event recognition and movement in hockey, enabling more precise insights into player performance. These advancements propel hockey science another step forward in the long-term goal of developing game-ready sensor-based HAR models that could be widely adopted by teams, equipment manufacturers, broadcasters, and other stakeholders. These innovations could streamline the process of annotating and analyzing movements, freeing up time for coaches and players to focus on strategic planning, training, and other essential aspects of the game.

3. Purpose, Objectives, and Hypotheses

The purpose of this study is to develop HAR hockey models using data that closely reflects real game conditions, while retaining enough experimental control to support the development of robust movement recognition algorithms. This study is designed to bridge the gap between a hockey environment that is completely controlled to an actual live game.

The objectives of this study are twofold.

1. To evaluate and compare the abilities of three deep learning architectures (a Transformer, a CNN, and a Hybrid model) in classifying eight ice hockey events using acceleration data collected in a game-simulated environment.
2. Compare the three models' performance using data from three different sensor configurations (17 sensors, 3 sensors, 2 sensors).

There are two hypotheses for the current study.

1. The hypothesis for the primary objective is that the Hybrid models will outperform the Transformer and CNN models, achieving a higher F1 score greater than 80% across all event types.
2. The hypothesis for the secondary objective is that the 17-sensor configuration will outperform the three and two-sensor configurations across all three model architectures, achieving a higher F1 score, and that the three-sensor configuration will outperform the two-sensor configuration.

4. Methods

4.1 Participants

Fifty-one ice hockey players, comprised of forty elite (male: n=20, female: n=20), and ten recreational (male: n = 8, female: n=3) players were recruited for this cross-sectional study. Of the fifty-one participants collected, forty-eight participant datasets were used in this study (Figure 1). The variability in the sample population was chosen to allow the final model to be more generalizable to a greater population, accounting for movement types of all players while favouring the more elite movements. Participants were selected via convenience sampling based on their willingness and eligibility to participate in the study. Participants were mainly recruited from the McGill University men's and women's varsity ice hockey teams. Others who fit the eligibility criteria were also recruited to participate in the study. "Elite" was defined as players who are fit for competition and have played in the Canadian Hockey League (Junior), the USports League (Canadian varsity university league), or in a league of an equivalent or higher calibre. Participants who played hockey below the previously mentioned "elite" levels were considered "recreational". Participants were excluded if they reported suffering from pain or injury during the time of the study that would prevent them from participating in their normal games or practices. Both left and right-handed shooters, as well as all skater positions (except goalies), were included. Players used their own sticks, skates, and gloves to encourage natural and comfortable performance. A full explanation of the required tasks for the study was given to the participants prior to signing an information and consent form in accordance with the Tri-Council Policy Statement on Ethical Conduct for Research Involving Humans. The Certificate of Ethics Approval was granted to the researcher on behalf of the McGill University Research Ethics Board on July 3rd, 2023 (McGill REB File #: 23-04-009) (Appendix A).

The sample size was based on a previous ice hockey event classification study, which used a similar sample size to obtain reliable results (Tremblay 2022). Furthermore, researchers have investigated annotated sample size effects on machine learning model performances and found that between 80 to 560 annotated samples were needed for consistent and accurate model performance (Figueroa et al., 2012). A systematic review on machine and deep learning for sport-specific movement recognition found that IMU-based studies (n=29) ranged from 1-30 participants (Cust et al., 2019). For the current study, a sample size of fifty-one participants exceeds many HAR studies in this field, generating an approximate minimum of 612 annotated samples per event type for the model to use for model development. This larger participant set is a strength of this study, as models will naturally generalize better if they are trained on more data (Chollet, 2018).

Table 1

Descriptive Statistics of Participant Data Included

	Total	Age (years) Mean ± SD	Height (cm) Mean ± SD	Weight (kg) Mean ± SD	Experience (years) Mean ± SD
Men	25	26 ± 6	181.4 ± 6.0	86.1 ± 9.8	20 ± 6
Women	23	21 ± 3	168.7 ± 5.0	64.0 ± 4.9	15± 3
Elite	40	23 ± 5	175.1 ± 8.2	75.5 ± 14.0	17 ± 5
Rec	8	27 ± 7	176.6 ± 10.1	75.4 ± 12.0	20 ± 9
All	48	24 ± 5	175.3 ± 8.5	75.5 ± 13.6	18 ± 6

4.2 Location

Data were collected at McGill McConnell Arena in Montreal, Quebec, on real ice from September 2023 through January 2024. The full ice surface was reserved for each data collection session. One participant was recorded per testing session, and one to three participants were recorded per testing day.

4.3 Instrumentation

4.3.1 Xsens MVN IMU System

Data were collected using the Xsens MVN Link (Xsens, Enschede, Netherlands) inertial measurement unit system recording at a sampling frequency of 240 Hz. This system was composed of 17 sensors, each containing a 3D linear accelerometer, a 3D gyroscope, and a 3D magnetometer. Sensors were connected in series using connective wires that fed into a communication terminal placed on the back of the participant in the pouch of a custom Xsens shirt (Figure 2). This terminal transmitted sensor data through a Wi-Fi signal to a router connected to a designated laptop that recorded the data. The wireless signal allowed the participant to skate around the entire ice surface freely without losing connection.

4.3.2 Video Recording

The entire testing session was recorded with a Dell WB70022c Ultra Sharp webcam (Dell Technologies, Austin, Texas, USA) recording at 60 Hz. It was connected to the same laptop recording the Xsens data and time synchronized with the Xsens recording software. Video recordings were used for event labelling during the post-hoc data pre-processing stage. The camera was set up on a stationary tripod in the stands of the arena, approximately 4 meters above and 5 meters away from the ice, in line with the home team's faceoff dots (Figure 3). The camera's field of view was wide enough to view all four nets so that the execution of each shot was captured.

4.4 Off-Ice Protocol

4.4.1. Participant Preparation and Information

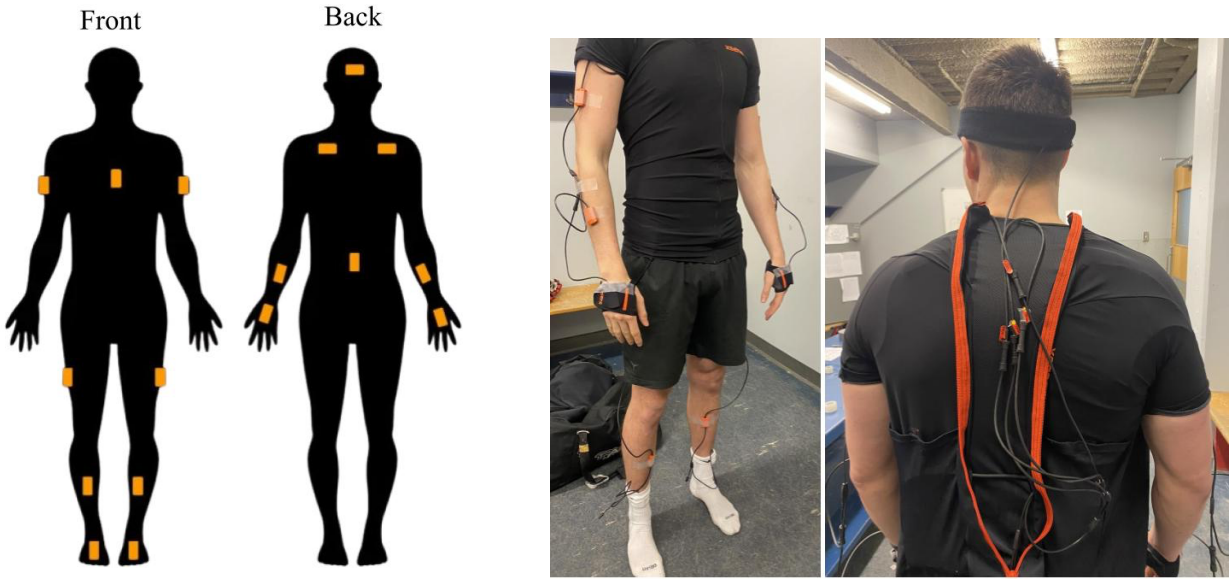
Participants were given a full explanation of the study on arrival before signing the required forms. The first form obtained their signed consent to participate in the study (Appendix B). The second form was a questionnaire for the participants to provide information about their playing experience, the equipment they use, and their injury history (Appendix C). Anthropomorphic measurements of each participant were then recorded using a measuring tape, and they were input into the Xsens software to generate the three-dimensional model of the participant. These measurements included total height, shoulder height, hip height, knee height, ankle height, foot length, foot width, hip width, shoulder width, elbow span, wrist span, and arm span. Participants' weights were recorded with an electronic scale.

4.4.2 IMU Placement

Next, the IMUs were placed on the participant's head, sternum, shoulders, upper arms, forearms, hands, pelvis, upper legs, lower legs, and feet (Figure 2). The head sensor was secured in a headband, and hand sensors were secured in slim gloves worn by the participants that fit inside their hockey gloves. The foot sensors were taped onto the tongue of the participants' skates. The rest of the sensors were secured with double-sided tape placed directly on the skin and reinforced with medical tape on top of each sensor to prevent sensor movement during testing. The connecting wires were taped down to the skin if needed and fastened inside pockets, holes, and zippers in the shirt that were designed to organize the wires.

Figure 2

IMU Sensor Placement on a Human Figure and Live Participants



4.5 On-Ice Protocol

4.5.1 Xsens Calibration

Once the sensors were fastened on the participants, they were escorted onto a carpet on the ice for the calibration trial. The calibration trial began with the participant standing in a static neutral pose before walking approximately four meters, turning 180 degrees, walking back to the starting point, and standing still in the static neutral pose again until the calibration trial was complete. After the calibration was completed, the participants had roughly five minutes to freely skate around the ice to warm up and become accustomed to the Xsens suit.

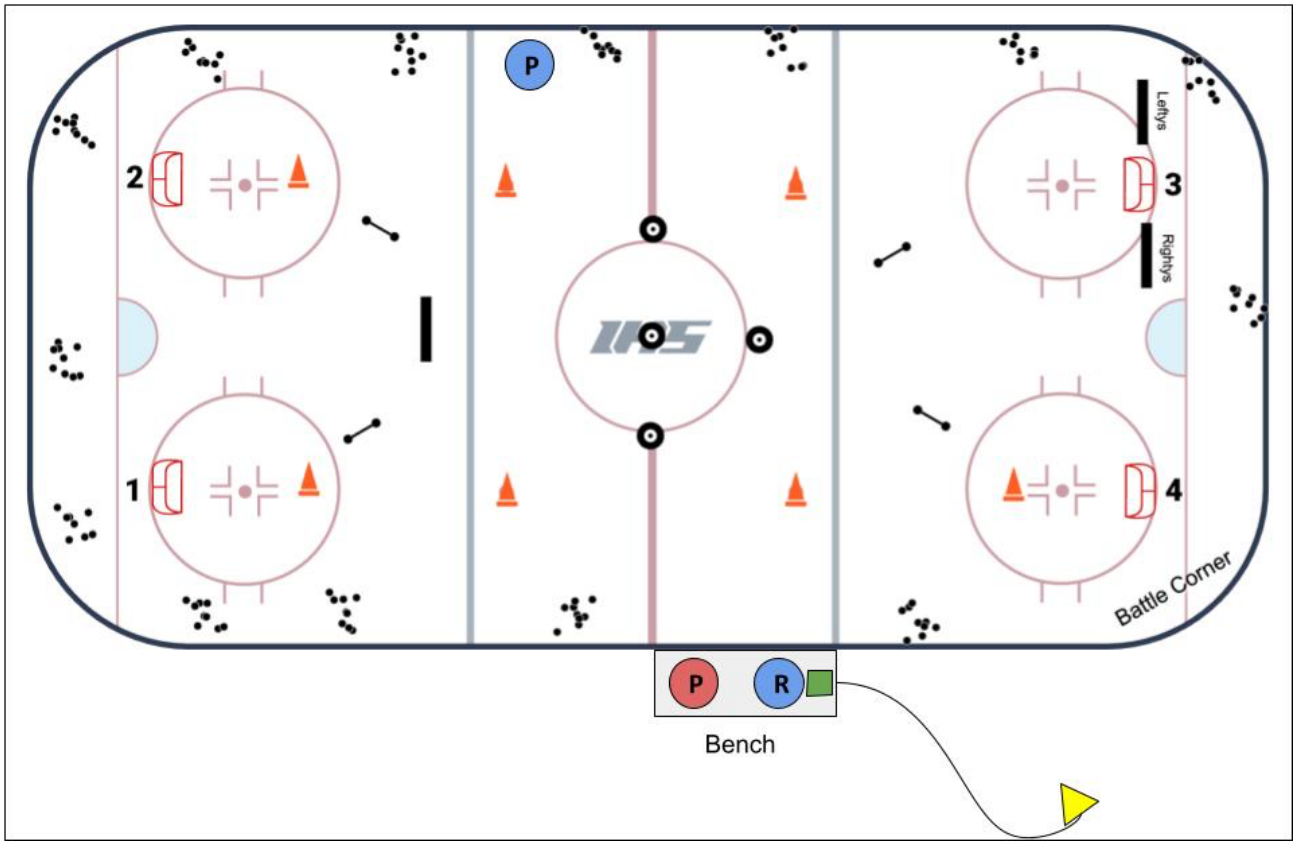
4.5.2 Obstacle Setup

One major focus of this study was to simulate the unpredictability and movement patterns that exist within a real hockey game. To achieve this, multiple obstacles were positioned on the ice in a way that forced the players to react quickly and adjust to their surroundings (Figure 3). Obstacles included pylons, stick-handling devices, and miniature wheels. The location of these

obstacles encouraged players to carry the puck and attack the net as they would in a real game, forcing them to set up and execute each shot differently. There were also two stationary self-passing aids for the participants to pass the puck to, which naturally rebounded the puck back to them (Give-n-Go Hockey, Michigan, USA). Pucks were scattered around the exterior of the ice, allowing the participant to receive pucks from different locations on the ice. Participants were instructed to perform the trial sequences at max effort within control, to treat the obstacles like opponents that they needed to beat, and to shoot as though they were trying to score on a goaltender, aiming for different spots of the net.

Figure 3

Diagram of the On-Ice Setup



Note. The red circle with the "P" indicates the starting position of the participant.
The blue circles with "P" represent the passer (The researcher on the ice distributing pucks to the participant).

The blue circle with the “R” on the bench is the researcher operating the laptop and giving commands.
The green square is the location of the computer setup.
The yellow triangle is the location of the camera.

4.5.3 Testing Protocol

Participants completed a series of fifty-five sequences broken up into eleven blocks. There were five sequences in each block, and each block was separated by a two-minute sitting rest period on the bench (Figure 4). The length of time to complete one block (roughly one minute) and the time resting on the bench (two minutes) were representative of a general average shift length and resting length that a player experiences in a real game.

A researcher on the bench verbally gave the three commands to the participant before each sequence, instructing them on 1) how they were to receive the puck (“pick” or “pass”), 2) the type of shot to perform (“slap shot”, “wrist shot”, “backhand”, or “one-timer”), and 3) which net to shoot at (“one”, “two”, “three”, or “four”). Once they completed a sequence, they skated back towards the bench to hear a researcher shout the next set of commands for the subsequent sequence. After completing five sequences in a row, the participant skated to the bench and rested for two minutes before completing another block of five sequences.

The participant received the puck in one of two ways. The first way was indicated by the command “pass” (n=25), in which the player received a pass from the passer. The passer was a researcher who assisted with the session on ice and was considered sufficiently skilled to support the study. The second method was indicated by the command “pick” (n=25). For a “pick”, the passer left a puck in a specific spot on the ice, and the participant simply skated to the puck and retrieved it themselves before continuing the sequence. The passer distributed the pucks from different areas of the ice to increase variability.

The protocol contained a relatively even distribution of shot types (Table 2). The order of reception method, the shot type, and the net were randomized for each participant, with the exception of *one-timer*s, which were always performed at net three to simplify the logistics of setting up for that shot type. The participants were instructed to perform at least one *pass* themselves per block against the stationary self-passing aids.

Figure 4

First Four Blocks (20 sequences) of an Example Protocol

Protocol - Example						
Shift/Block	**Sequence**	**Reception**	**Shot**	**Net**	**Self Pass?**	**Start/End Time**
1	**1**	pick	Back Hand	1		
	2	pick	One Timer	3		
	3	pick	Back Hand	1		
	4	pass	Wrist	1		
	5	pass	Back Hand	1		
Rest						
2	**6**	pass	Wrist	2		
	7	pick	Back Hand	4		
	8	pick	Slap	2		
	9	pick	Wrist	1		
	10	pick	Slap	1		
Rest						
3	**11**	Drill/Battle	Any	4		
	12	Drill/Battle	Any	4		
	13	Drill/Battle	Any	4		
	14	Drill/Battle	Any	4		
	15	Drill/Battle	Any	4		
Rest						
4	**16**	pick	Wrist	2		
	17	pass	Back Hand	1		
	18	pass	Back Hand	4		
	19	pass	Wrist	2		
	20	pick	One timer	3		
Rest						

Table 2

Distribution of Shot Commands for Each Participant

Shot Command	Total
Slap Shot	13
Wrist Shot	13
Back Hand	12
One-Timer	12
Participant Choice	5

An example command for a sequence that a participant could have heard is “Pass. Slap. One”. This meant that they would receive a pass from the passer and carry the puck through the obstacles towards net one to perform a *slap shot*, then skate back to the bench for another set of commands or *rest*. Another example could be “Pick. Wrist. Three”. In this case, the player would retrieve the designated puck set out by the passer, deke around the obstacles, perform a *wrist shot* on net three, and then skate back to the bench for another set of commands or *rest*.

The setup and execution for the *one-timer* were unique compared to the other shots. If the command was “Pick. One-timer. Three”, the participant would pick up a puck, maneuver through the obstacles to net three, pass their puck to the self-passing aid beside the net, and perform their *one-timer* from the rebounded pass. If the command was instead “Pass. One-timer. Three”, the participant would simply skate directly to net three to receive a pass from the passer and perform their shot from that pass. In either scenario, they would then carry on to their next sequence.

To further create a realistic environment, one of the eleven blocks contained two drill sequences and three battle sequences, which were completed with the assistance of the passer. The drill sequences, performed on both sides of the net four, involved a give-and-go passing drill before shooting on net four with a shot of their choice. In the battle sequences, the participant

had to maintain control of the puck in the corner while the passer applied pressure to them and tried to take the puck from them. After roughly seven seconds of battling for the puck in the corner, the participant curled in front of the net and executed a shot of their choosing.

Given the nature of the study and the requirements for training the deep-learning algorithm, if a player made a mistake, like losing the puck, performing the wrong shot, or shooting at the wrong net, the error was not corrected. These errors further represent the unpredictable nature of ice hockey, and having slightly more or less of a shot type for a participant would not affect the study, so long as the general shot distribution remained relatively uniform across all participants.

4.6 Data Processing

4.6.1 Event Labelling

The time-synchronized video footage was used as ground-truth data to label when each event occurred within the data collection session. These event labels were essential for the input data that trained the deep learning model. Using video annotating software (ELAN, Nijmegen, Netherlands), two researchers recorded the video's time stamps for each event to the thousandth of a second. The beginning of each shot sequence to the end of the player's follow-through was labelled either "WS" (wrist shot), "SS" (slap shot), "BH" (backhand), "OT" (one-timer), or "Pass" (pass). Frames in which the player was skating without the puck (i.e., before they received a puck or after a shot) received the label "NP" (no puck). Frames that featured the player skating in possession of the puck received the label "PC" (puck control). Frames, where the player was on the bench resting, received the label "Rest". Frames before the official trial began, and after the trial was completed, as well as frames where the participant was out of frame or performed an action that is not representative of hockey, received the label "donotuse"

and thus were excluded from the final dataset. A total of 9661 events were recorded and annotated (Table 3).

To assign the labels from the videos to the participant's acceleration data, the Xsens system needed to first reprocess and export the data using the Xsens MVN Analyze software. Only the sensor-free acceleration data (i.e., the acceleration detected by each IMU sensor without the Earth's gravity acceleration component) (Xsens, 2018) were used. Only acceleration data were used to reduce model complexity and reduce the computational load, while still providing a detailed signal capable of capturing the dynamic movements characteristic of hockey. This approach also aligns with prior work in this field (Hardegger et al., 2015; Tremblay, 2022). The specific timestamp for each event from the annotation software was then multiplied by 240 to correspond with the Xsens capture rate (240Hz) and added to the exported acceleration data so that each frame of acceleration could, therefore, have a corresponding label. There were roughly 220 labelled events for each participant. The labelled acceleration data were then used in the deep learning model development and evaluation stages.

4.6.2 Data Preprocessing

Each participant's data were roughly 500,000 frames because of the continuous recording of the entire session. Of the 51 participant trials that were collected, 3 had to be excluded due to file corruption or other technical issues. As a result, the remaining 48 participant trials were uploaded to a Google Drive (Google LLC, Mountain View, US) for storage and use. All preprocessing and model development steps were completed using Python software (Python Software Foundation, https://www.pyton.org/) on Google's Colaboratory Pro+ GPU (GPU: NVIDIA A100-SXM4-40GB).

The first preprocessing step was to convert all the right-handed participants to reflect left-handed ones to ensure consistency among all the data. To do this, sensor column headings that had the word "right" (arm and leg sensors) were switched with the word "left" and vice versa. For the columns that did not have the word "left" or "right" (head, sternum, and sacrum sensors), a rotation matrix was applied to rotate the data 180 degrees around the z-axis. After all the participants' accelerometer data reflected left-handed shooters, the separate participants' data frames were concatenated together into one large data frame. Each corresponding event label (e.g., *wrist shot, backhand, pass,* etc.) was then encoded into a unique integer, a requirement of deep learning models (Chollet, 2018). The labels were later one-hot encoded into a binary vector representation (e.g., *backhand* was represented as [0, 0, 1, 0, 0, 0, 0, 0]). This is the suitable format for multiclass classification.

From here, the data could be separated into three different sensor configurations. The first one, being the all-sensor configuration (ASC), included acceleration data from all 17 sensors, for a total of 51 channels (17 sensors * 3 axes). The second configuration was the sternum and hands sensor configuration (SHSC), which included 3 sensors, resulting in 9 channels. The final configuration was the hands-only configuration (HSC), which included the two hand sensors, resulting in 6 channels. These three sensor configurations were selected to evaluate the trade-off between model performance and sensor practicality. The ASC could provide complete biomechanical coverage and act as the gold standard reference. The SHSC represents a reduced setup that still captures both central and distal limb motion relevant to upper body actions. The HSC was included to test the feasibility of a minimal setup, as it provides a potential solution that interferes less with a player's natural movements and could be more practical for live in-game deployment. These configurations are also consistent with previous hockey HAR research

(Hardegger et al., 2015; Tremblay, 2022). The following preprocessing steps were repeated for each sensor configuration.

The deep learning models in this study require a uniform shape for each sample of input data (acceleration sequence). Therefore, shorter samples were rear-padded to a target length, while longer samples were truncated to match the same length (Chollet, 2018). In this case, the target length was determined by the longest shooting sample (*one-timer*), which consisted of 664 frames (2.77 seconds). However, to minimize the number of samples with excessive padding, samples that were shorter than 150 frames (0.625 seconds) were excluded and removed from the data set (Table 4). This approach was chosen to preserve the integrity of the shooting event signals, which were prioritized over other events such as *puck control, no puck, pass, and rest*.

Table 3

Total Number of Available Samples for Each Event

Event Type	Count
Slap Shot	601
Wrist Shot	765
Back Hand	619
One-Timer	561
Pass	738
Puck Control	2497
No Puck	3379
Rest	501

Table 4

Number of Samples Excluded Due to Being Shorter than 150 Frames

Event Type	Count
Puck Control	47
Pass	44
No Puck	22
Back Hand	17
Wrist Shot	13
One-Timer	1

Additional steps were required to ensure the data were representative and adequately prepared for model development. To prevent bias in the model, a balanced subset of 560 samples of each event type was randomly selected for inclusion. This number was chosen based on the *one-timer* events only having 560 samples total, which was the fewest of all stick-based events. Although there were only 501 samples of rest, it was determined that because of the unique characteristics of the resting action (e.g., sitting stationary on the bench) compared to the other events, having fewer rest samples would not negatively impact the model's performance. The seed to randomly select the 560 samples of each event remained consistent across all three configurations to ensure consistency amongst datasets.

The next step was to split the dataset into training, validation, and test tensors. The train-test split was done participant-wise to ensure that no participant had data in more than one dataset. This approach helps to avoid data leakage and makes the model more generalizable. The randomized split assigned 70% of the participants to the training set (n=33), 10% to the validation set (n=5), and 20% to the testing set (n=10). The number of males and females, as well as the number of elite and recreational participants, was randomized for the training, validation, and testing tensors. The training set was used to train the model, the validation set was used to monitor the model during hyperparameter tuning and training, and the test set was saved until the end to evaluate the trained model's performance. The seed to randomly split the participants also remained consistent.

The final preprocessing step involved standardizing the amplitude of the datasets. Some models are very sensitive to scale and might struggle with the vastly different ranges that can commonly exist in 3-dimensional acceleration data. To address this, each sample in the input data was standardized by subtracting the mean and dividing by the standard deviation, resulting

in the data being centred around 0 with a standard deviation of 1 (Chollet, 2018). Using the Scikit-learn library (Pedregosa et al., 2011), a Standard Scaler was fit exclusively to the training set and subsequently applied to transform the training set, validation set, and test set.

As a result of all the preprocessing steps, the final shape of the training data tensors was *n* x 664 x *f*, where *n* represents the number of samples, 664 is the length of each sample, and *f* represents the number of channels. For example, the tensor shape for the accelerometer training data for the ASC was (3473, 664, 51), and the corresponding label tensor shape was (3473, 8).

4.6.3 Model Development

4.6.3.1.Model Architecture

The deep learning models were developed using the TensorFlow library (Google, 2023). Three different deep learning neural networks were developed and compared: a Transformer Network (Figure 5), a Convolutional Neural Network (Figure 6), and a Hybrid network containing both Convolutional and Transformer blocks (Figure 7). To ensure consistency across models, the architecture remained similar for all three configurations while adjusting the hyperparameters for each specific model. In the end, nine models (3 architectures * 3 sensor configurations) were developed and evaluated on their ability to classify the eight different hockey events from an unseen test set. To be concise, the following section will only describe the Hybrid model architecture, as it incorporates both the CNN and Transformer components within it. The corresponding code can be found on the private investigator's GitHub Repository dedicated to this project.

The first section of the hybrid model contains two one-dimensional CNN blocks (Equation 1). CNNs are effective at local feature extraction in complex accelerometer data segments (Deep & Zheng, 2019). They also reduce the dimensionality of the input data, which

can reduce the high computational demands of the Transformer. These blocks begin with convolutional layers with an L2 regularizer. These regularizers act as a penalty term in the loss function and are implemented to mitigate overfitting by forcing the weights to take smaller values, thus making the model less sensitive to noise in the data. This is followed by batch normalization, which helps stabilize the training by normalizing the outputs of the convolutional layers. Next, a rectified linear unit (ReLU) activation function is applied to introduce non-linearity, enabling the model to learn complex patterns by keeping positive inputs unchanged and turning negative inputs to zero. Then, it uses a MaxPooling layer to downsample the sequence, further reducing the dimensionality. Finally, a dropout layer is used to reduce overfitting by randomly deactivating a fraction of the neurons during training (Chollet, 2018). The layers following the second convolutional block connect the output of the CNN to the Transformer using a dense layer to project the outputs to a higher-dimensional space, matching the dimensionality requirements of the Transformer.

This data then passes through two components of the Transformer block. The first is the scaled dot-product attention, which calculates the attention scores between tokens (Equation 2). This model uses multiple "attention" heads, which simultaneously allows the model to capture different types of relationships by concatenating the projected scaled dot-product attention scores (Equation 3). Using multiple attention heads is advantageous as different contexts observed from the different heads can be combined into a single complex output (Dirgová Luptáková et al., 2022). The multi-head self-attention block also features dropout, L2 regularizer, and normalization layers. The second block in the Transformer is the feed-forward network, with two dense layers with a ReLU activation function between them (Equation 4). Both blocks contain layer normalization, dropout, L2 regularization, and residual connections, which are shortcuts

that allow a layer's inputs to bypass certain layers if they are unnecessary and skip directly to the output of the given layer (Equation 5). This improves gradient flow and stabilizes the training.

The final section of the model is the classification head. In this section, the sequence-level features are converted into a class for final prediction. This section contains a Global Average Pooling Layer to reduce the sequences to a fixed-size vector, a final dropout layer, and concludes with a dense (fully connected) layer with an L2 regularizer and SoftMax Activation Function, which generates a probability for each of the eight classes.

Equation 1: Multi-channel 1D CNN

$$y_i^{(f)} = \sum_{c=1}^{C} \sum_{j=0}^{k-1} x_{i+j}^{(c)} * w_j^{(f,c)} + b^{(f)}$$

Equation 2: Attention

$$Attention\ (Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k)\ V}}\right)$$

Equation 3: Multi-Head Attention

$$MultiHead\ (Q, K, V) = Concat(head_1, \ldots, head_h)W^{o,}$$

Where $head_i = Attention(WQ_i^Q, KW_i^K, VW_i^V)$

Here, WQ_i^Q, KW_i^K, and, VW_i^V are the projection matrices for each head, and W^o is the output projection matrix.

Equation 4: Feed Forward Network

$$FFN(x) = ReLU(xW_1 + b_1)W_2 + b_2$$

Equation 5: Output Layer

$$Output = LayerNorm\ (x + Sublayer(x))$$

Figure 5

Schematic of the Transformer Model Architecture

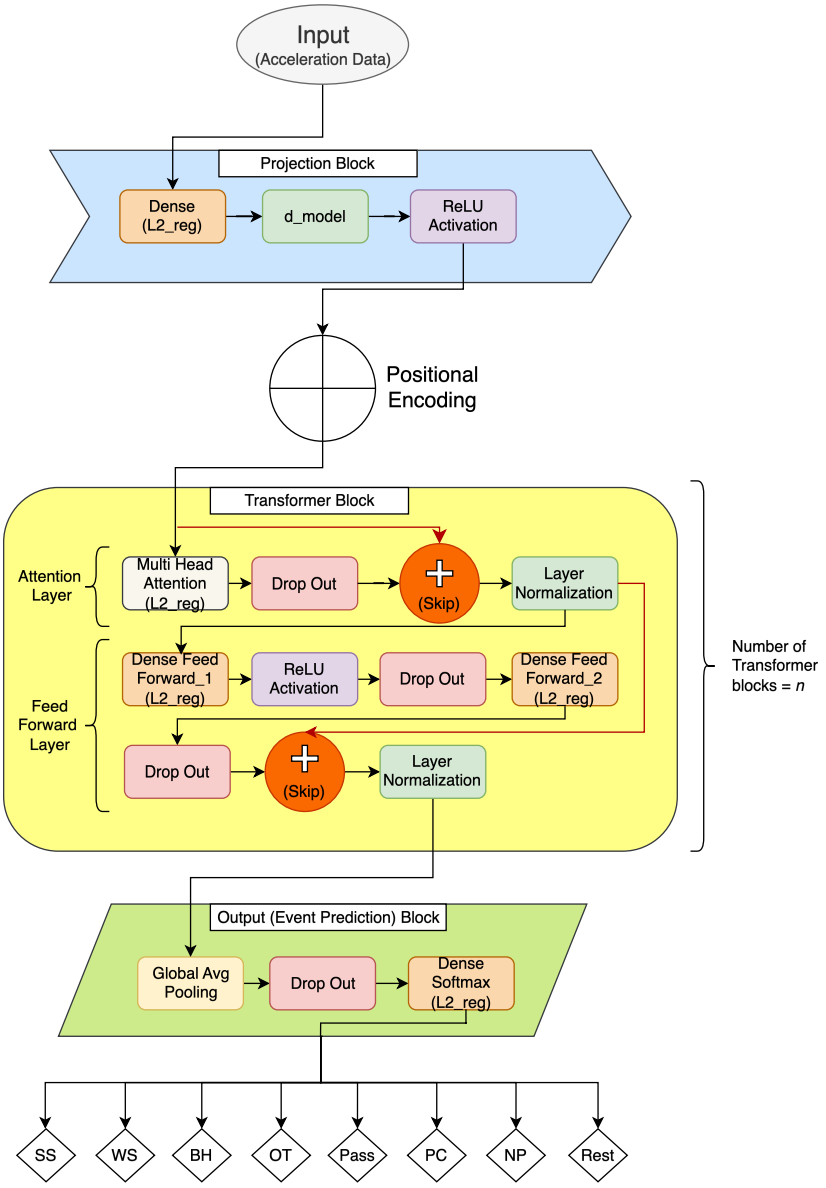


Figure 6

Schematic of the CNN Model Architecture

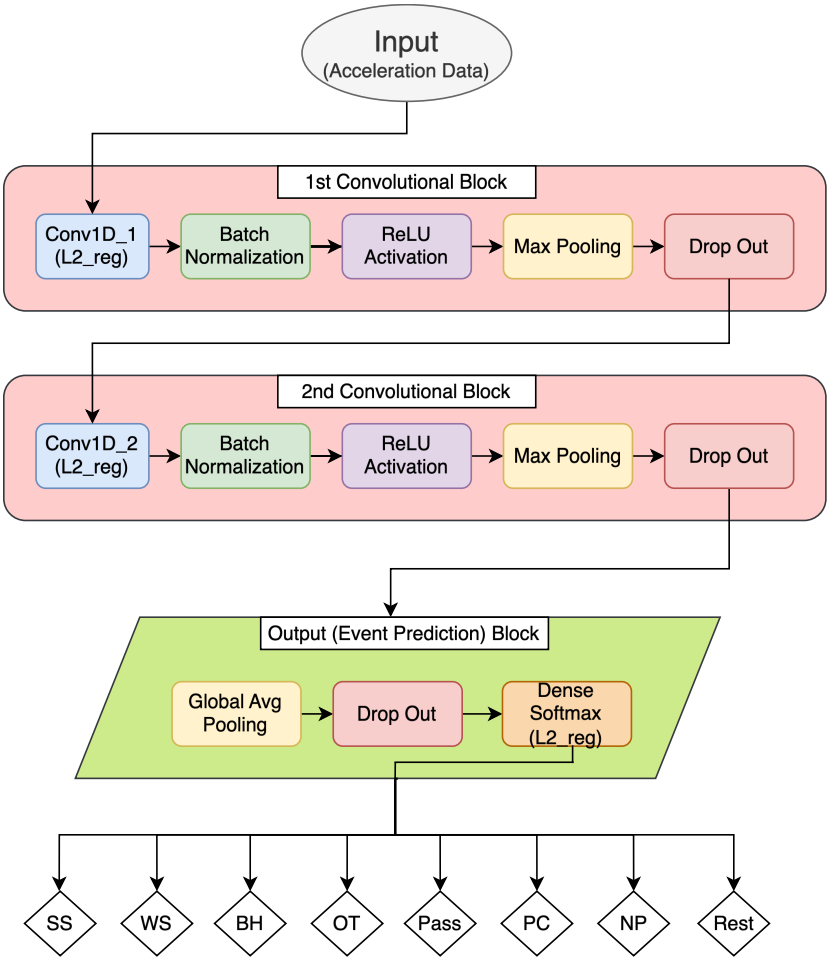
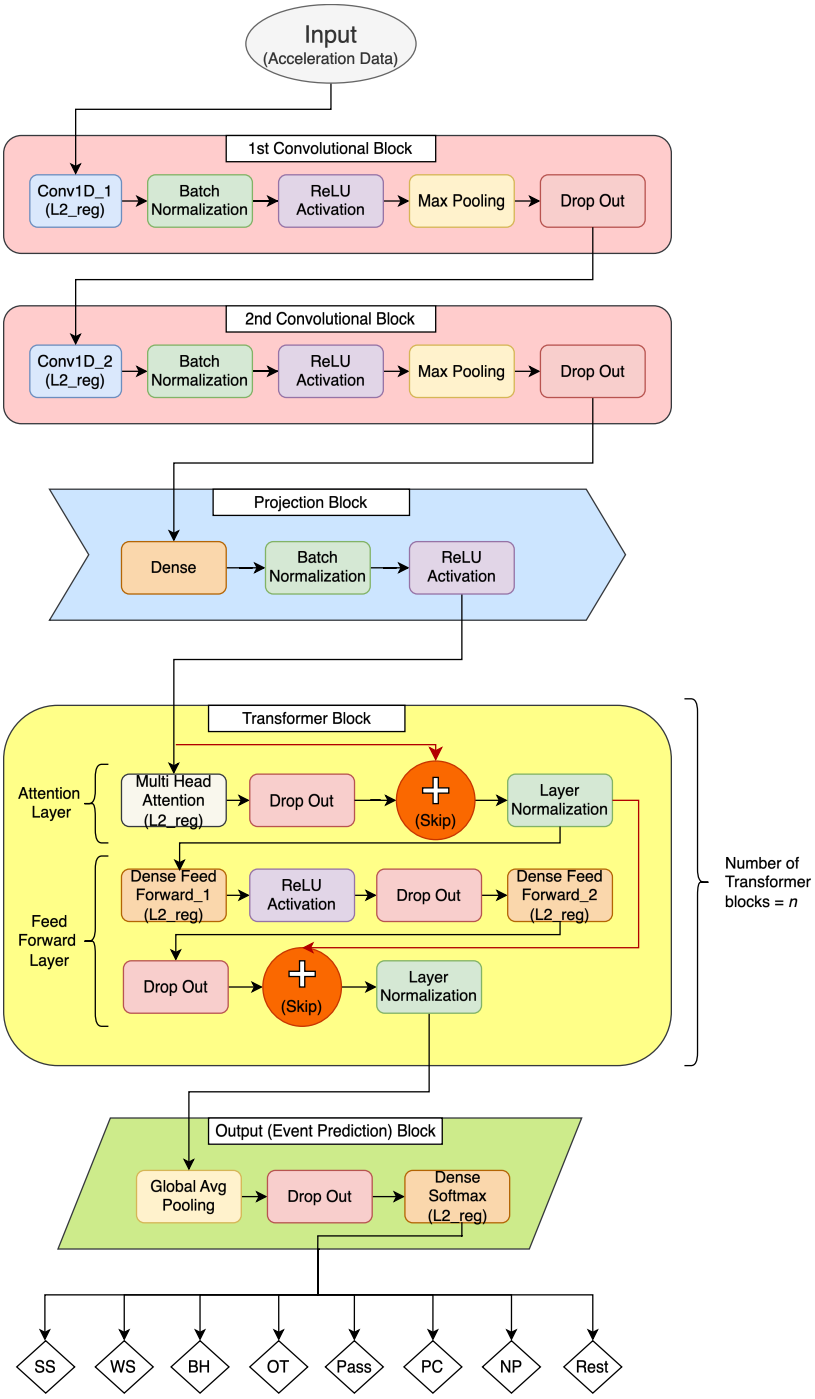


Figure 7

Schematic of the Hybrid Model Architecture



4.6.3.2 Transformer Positional Encoding

The Transformer models in this study contain a unique step that the CNN and hybrid models do not. When processing an acceleration signal, Transformers treat each individual timestep as a token. These tokens are embedded in parallel, meaning they do not possess the inherent sequential processing capabilities of recurrent or convolutional models. Therefore, it is important to inject positional information into each token so that the model can recognize the important sequential patterns in the data. To address this, *positional encodings* can be implemented before the encoder, maintaining the same model dimension (*model*) as the embeddings. The two main choices for this are fixed or learned positional encodings. Learned encodings are tuned as a parameter, but once tuned, they are limited to the learned fixed lengths. For the given study, the fixed positional encoding method was chosen using sin and cosine functions, as they can allow for longer sequence lengths than the ones they were exposed to in training. The different frequencies of interleaving sine functions for even dimensions (Equation 6) and cosine functions for odd dimensions (Equation 7) provide each sequence with a unique encoding, and the relative differences in the positional encodings allow the model to compute their unique relationships. The formulas and methods used in this study were first introduced in the paper "Attention is All You Need" (Vaswani et al., 2023). The CNN and Hybrid model do not need this step because the convolutional layers inherently capture order and relationships through their receptive fields, essentially generating positional context in the extracted features.

Equation 6: Even Positional Encoding (sin)

$$PE_{(pos,2i)} = \sin\left(pos/1000^{2i/dmodel}\right)$$

Equation 7: Odd Positional Encoding (cosine)

$$PE_{(pos,2i+1)} = \cos\left(pos/1000^{2i/dmodel}\right)$$

4.6.3.3 Hyperparameter Tuning

To optimize the model's performance, several hyperparameters were tuned using Optuna (Akiba et al., 2019). Optuna's default parameter tuning method, and the one used in the current study, is the Tree-Structured Parzen Estimator. This method uses Bayesian optimization to model the probability of good and bad hyperparameters based on previous trials to suggest hyperparameters that will likely optimize the model's performance. Tree-Structured Parzen Estimator has advantages over standard tuning methods like Random Search or Grid Search because it focuses on the promising region of the parameter space and adapts its tuning based on past trials, generating a more efficient and targeted search (Figure 8). A range of hyperparameters were explored for model optimization (e.g., learning rate, batch size, dropout rate, regularization coefficient), CNN architecture (e.g., number of filters, kernel size), and Transformer architecture (e.g., number of transformer layers, model dimension, feed-forward dimension, number of attention heads) (Figure 9). For each model, Optuna ran 20 trials for 20 epochs using the training dataset, and the best-performing hyperparameters were selected and used in the final model build. While hyperparameter importance varied slightly, the learning rate was often observed to be the most important hyperparameter across all nine models (Figure 10).

Figure 8

Parameter Optimization History Plot for SHSC CNN Model

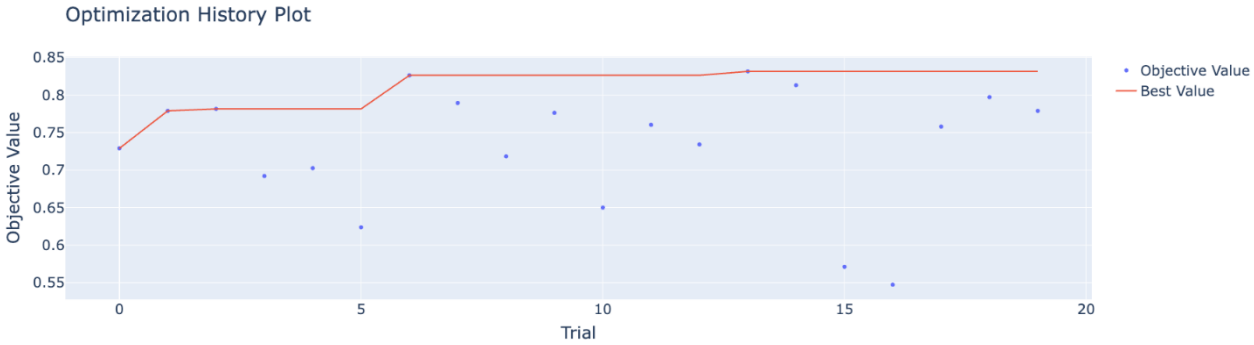


Figure 9

Parameter Optimization Parallel Coordinate Plot for HSC Hybrid Model

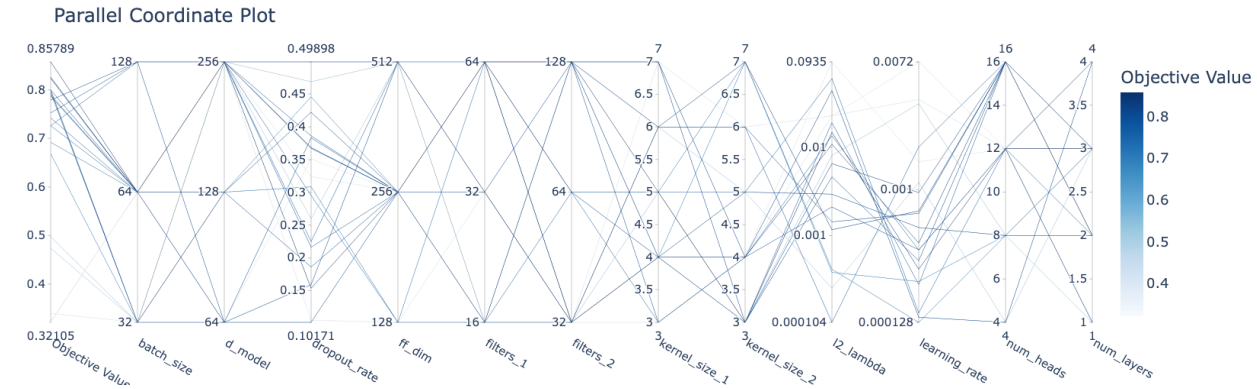
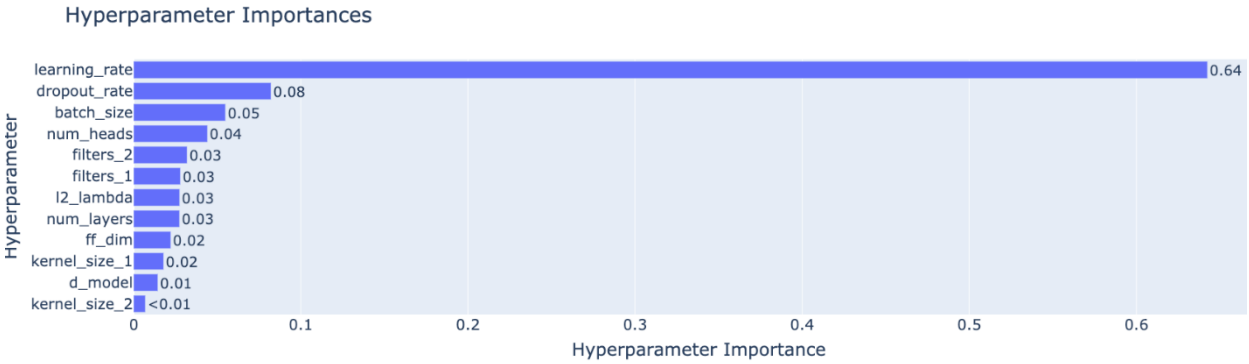


Figure 10

Parameter Tuning Importance Plot for HSC Hybrid Model



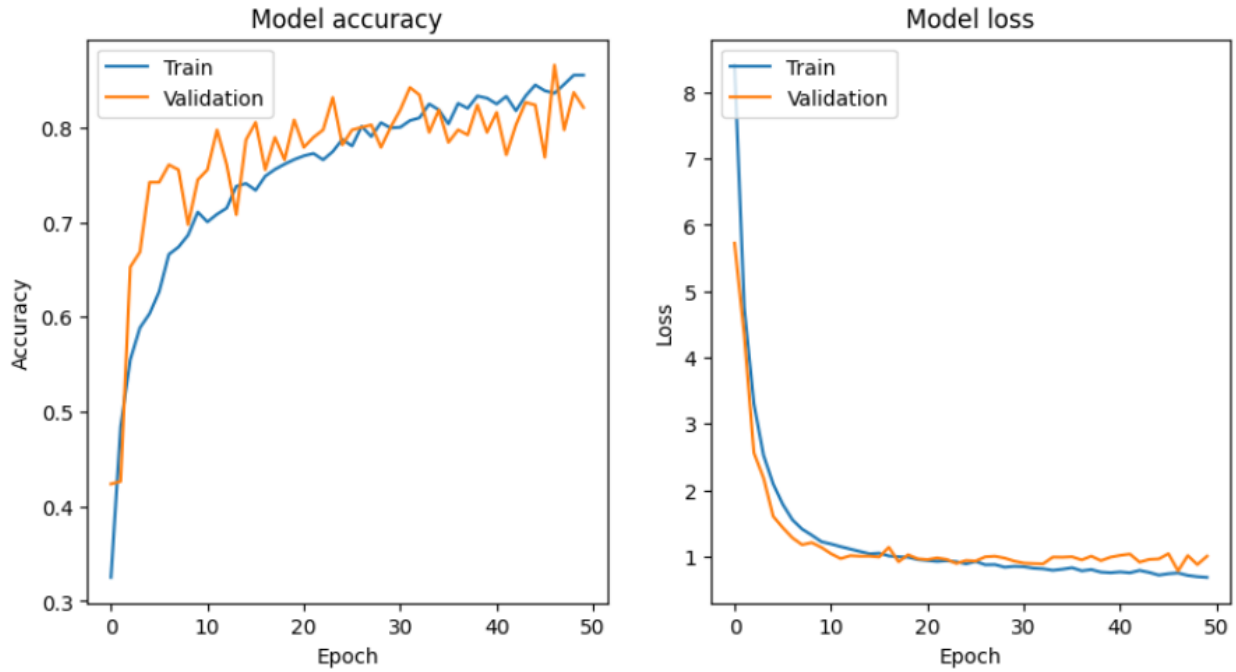
4.6.3.4 Model Compilation

The final section in preparing the model for training was the compilation step. This stage includes a loss function to measure the model's performance on the training data, an optimizer to update the model's weights and biases based on data from the loss function, and a metric to monitor the training (Chollet, 2018). The chosen loss function was categorical cross-entropy, which calculates the difference between the predicted probability distribution from the SoftMax output to the true label. Adam optimizer was used, which stands for "adaptive moment estimation". The Adam optimizer adapts its learning rate for each parameter based on the mean

and the variance of the gradients during training. The evaluation metric used was accuracy, which is based on the number of correctly classified samples out of the total number of samples. The same loss function, optimizer, and evaluation metric were used for each of the nine models built in this study. Each model was trained for 50 epochs, and the loss curves were closely monitored for overfitting (Figure 11).

Figure 11

Accuracy and Loss Curves During Training for ASC Transformer Model



4.7 Statistical Analysis

The nine fully trained models were evaluated using the same test set, generating an F1 score to measure each model's performance. The F1 score (Equation 8) is a balanced measure of a model's accuracy, comprised of a weighted average of precision and recall scores (Mishra, 2018). F1 scores range from zero to one, with a higher score indicating better model performance. Precision (Equation 9) and Recall (Equation 10) were also used to measure the

models' overall performance and ability to classify each specific event. Precision is the proportion of correctly predicted positive samples, indicating the reliability of those predictions. For example, precision tells us how many of the predicted slapshots were actually slapshots. Recall is the proportion of actual positives correctly identified. It tells us that of all the slapshots that exist, how many of them did the model detect?

Equation 8: F1 Score

$$F1\ score = \frac{2\ x\ (Precision\ x\ Recall)}{Precision + Recall}$$

Equation 9: Precision

$$Precision = \frac{True\ Positive}{True\ Positive + True\ False\ Positive}$$

Equation 10: Recall

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

5. Results

All the models performed well, with F1 scores ranging from 72.2% to 83.7% (Table 5). The Hybrid model architecture built with the sternum and hands sensor configuration achieved the highest F1 score of 83.7% (Table 6).

Table 5

Summary of Results

Model Type	Sensor Config.	F1 Score	Mean F1 Score	Precision	Mean Precision	Recall	Mean Recall
Transformer	ASC	77.2		77.8		77.3	
	SHSC	73.8	74.4	75.2	75.3	73.8	74.5
	HSC	72.2		73.6		72.4	
CNN	ASC	81.8		82.8		82.0	
	SHSC	79.8	79.5	81.2	80.0	80.2	79.7
	HSC	76.9		79.1		77.2	
Hybrid	ASC	81.6		82.1		81.7	
	SHSC	83.7	81.9	84.6	83.5	83.6	82.2
	HSC	80.4		83.7		81.2	

Note.
ASC = All Sensor Configuration
SHSC = Sternum and Hands Sensor Configuration
HSC = Hands Sensor Configuration
CNN = Convolutional Neural Network

Table 6

Classification Report for SHSC Hybrid Model (Highest Performing Individual Model)

Event	Precision	Recall	F1 Score
Slap shot	68.1	81.7	74.3
Wrist shot	87.6	80.7	84
Back hand	97.9	78	86.8
One-timer	69.9	69.9	69.9
Pass	77.3	93.5	84.7
Puck Control	91.2	83.2	87
No Puck	91.7	85.3	88.4
Rest	93.7	97.2	95.4
Weighted avg.	84.6	83.6	83.7

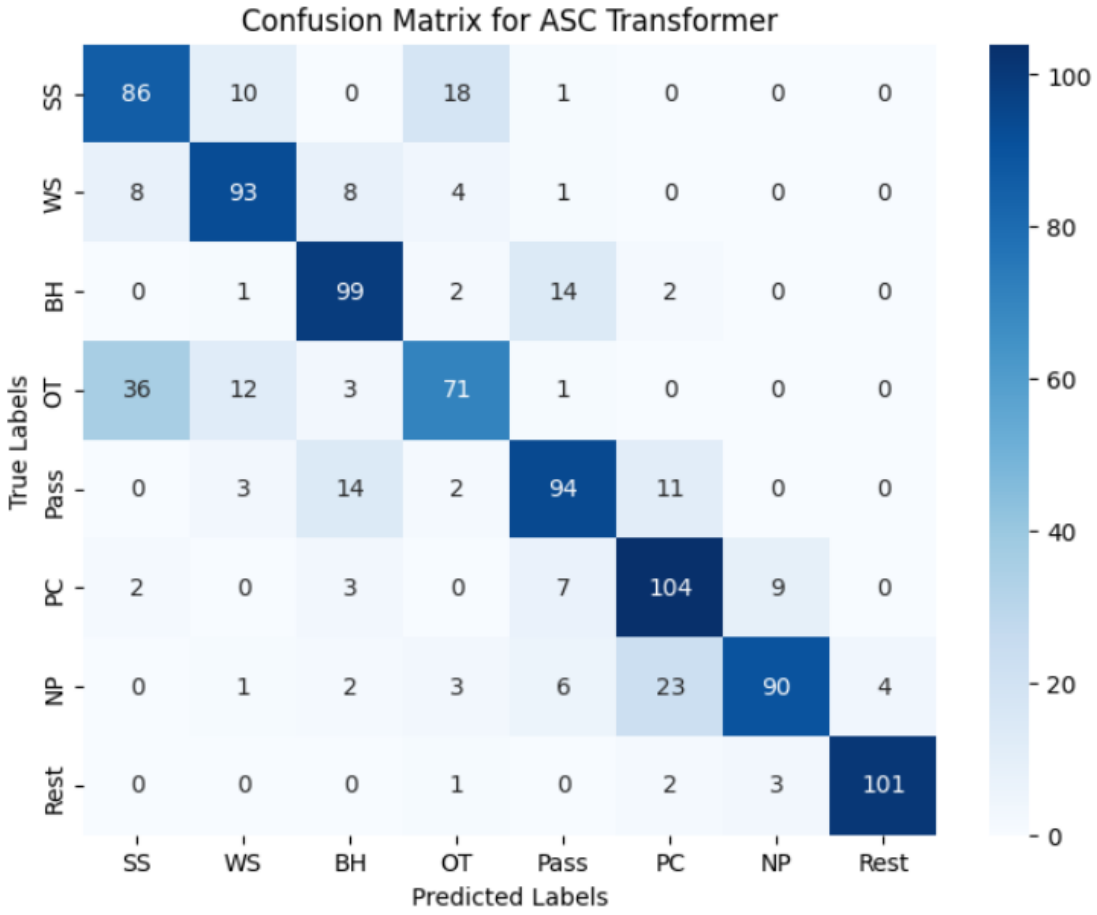
5.1 Transformer Model Performance

The transformer model performed well overall, but it encountered the most difficulty out of the three model types (Table 5). The F1 score for the ASC was 77.2%, demonstrating the highest F1 score for the *rest* events (95.3%) and the lowest F1 score for the *one-timer* events (63.4%). The F1 score for the SHSC was 73.8, demonstrating the highest F1 score for the *rest* events (90.8%) and the lowest for the *one-timer* events (63.6%). The F1 score for the HSC was 72.2%, demonstrating the highest F1 score for the *rest* events (88.7%) and the lowest F1 score for the *one-timer* events (62.2%). Confusion matrices were generated for all three configurations, revealing the number of predicted labels vs the true labels (Figure 12).

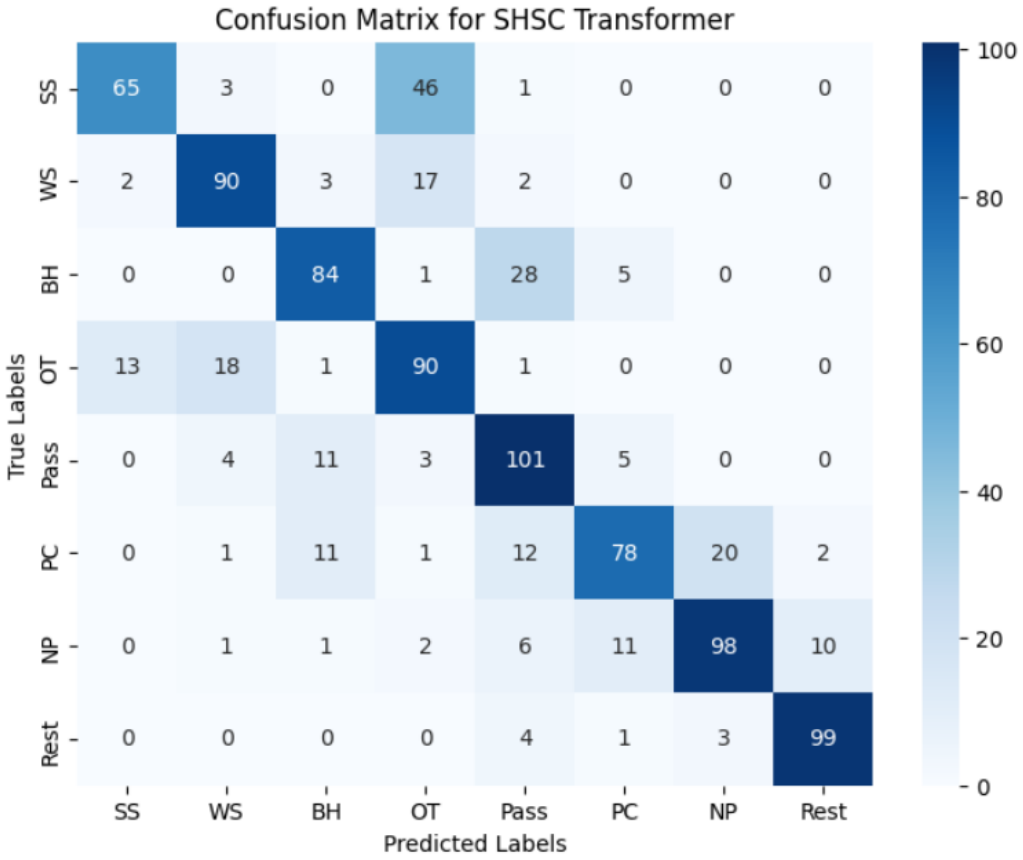
Figure 12

Confusion Matrices for the Transformer Models with the ASC (a), the SHSC (b), and the HSC (c)

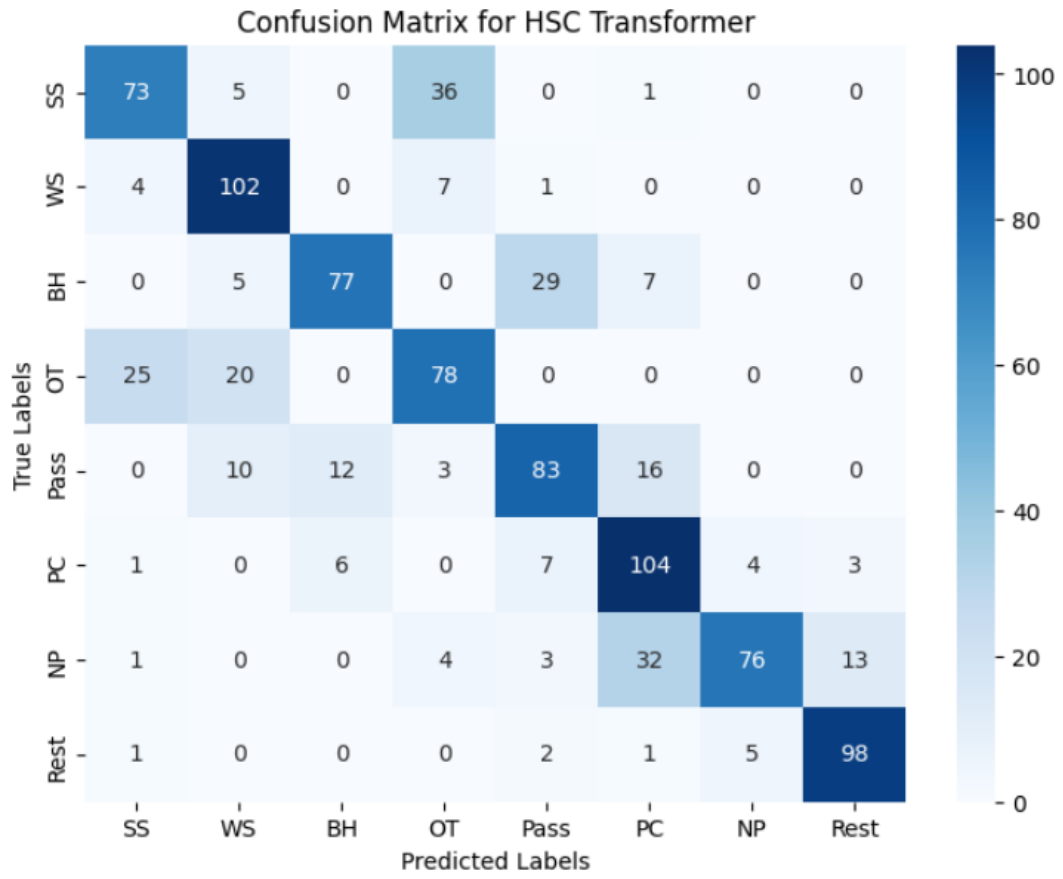
a)



b)


Confusion Matrix for SHSC Transformer
True Labels
Predicted Labels
SS
WS
BH
OT
Pass
PC
NP
Rest
65 3 0 46 1 0 0 0
2 90 3 17 2 0 0 0
0 0 84 1 28 5 0 0
13 18 1 90 1 0 0 0
0 4 11 3 101 5 0 0
0 1 11 1 12 78 20 2
0 1 1 2 6 11 98 10
0 0 0 0 4 1 3 99
100
80
60
40
20
0

c)



Note. HSC = Hands Sensor Configuration; SS = Slap Shot; WS = Wrist Shot; BH = Back Hand; OT = One-Timer; PC = Puck Control; NP = No Puck

5.2 CNN Model Performance

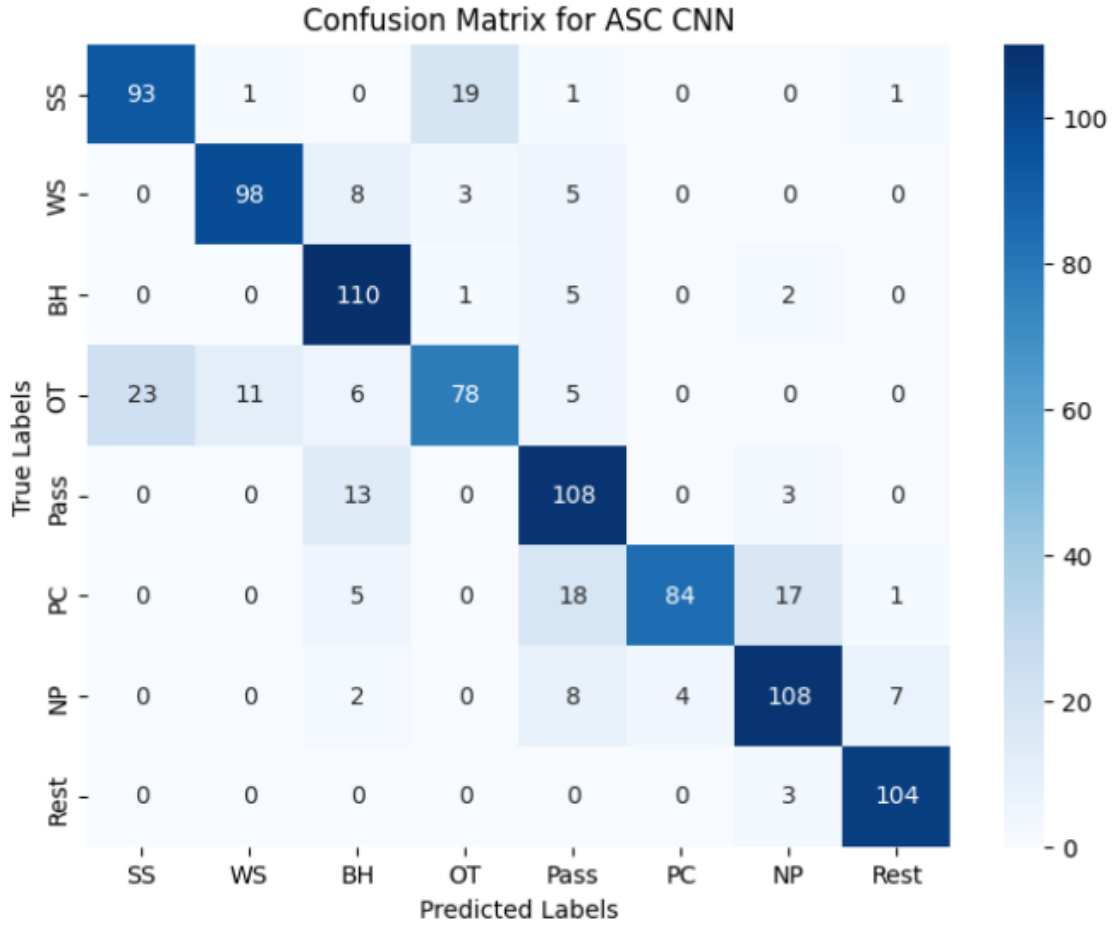
Overall, the CNN performed well across all three sensor configurations (Table 5). The ASC achieved an F1 score of 81.8%, demonstrating the highest F1 score for the *rest* events (94.5%), and the lowest for the *one-timer* events (69.6%). For the SHSC, the CNN achieved an F1 score of 79.8%, demonstrating the highest F1 score for the *rest* events (93.1%) and the lowest for the *one-timer* events (55.6%). For the HSC, the CNN achieved an F1 score of 77%, demonstrating the highest F1 score for the *rest* events (92%) and the lowest for the *one-timer*

events (61.3%). Confusion matrices were generated for all three configurations, revealing the number of predicted labels vs the true labels (Figure 13).

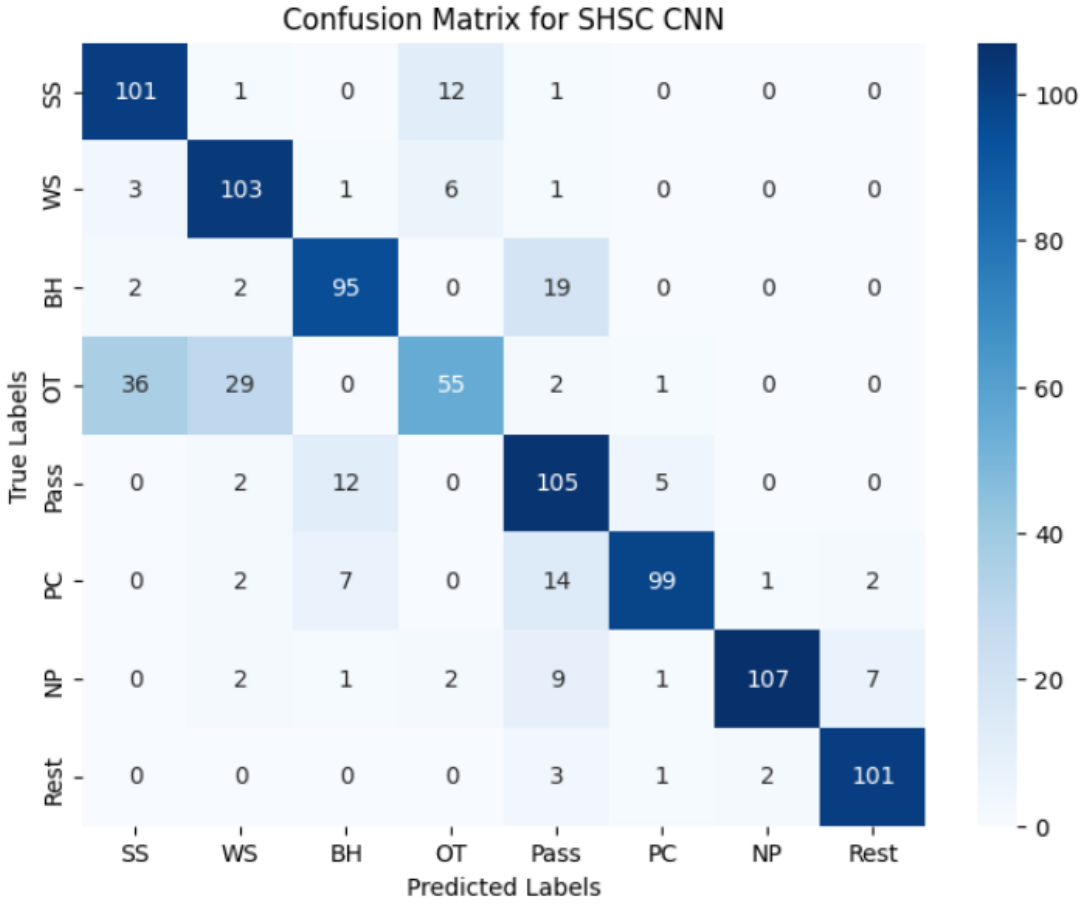
Figure 13

Confusion Matrices for the CNN Models with the ASC (a), the SHSC (b), and the HSC (c)

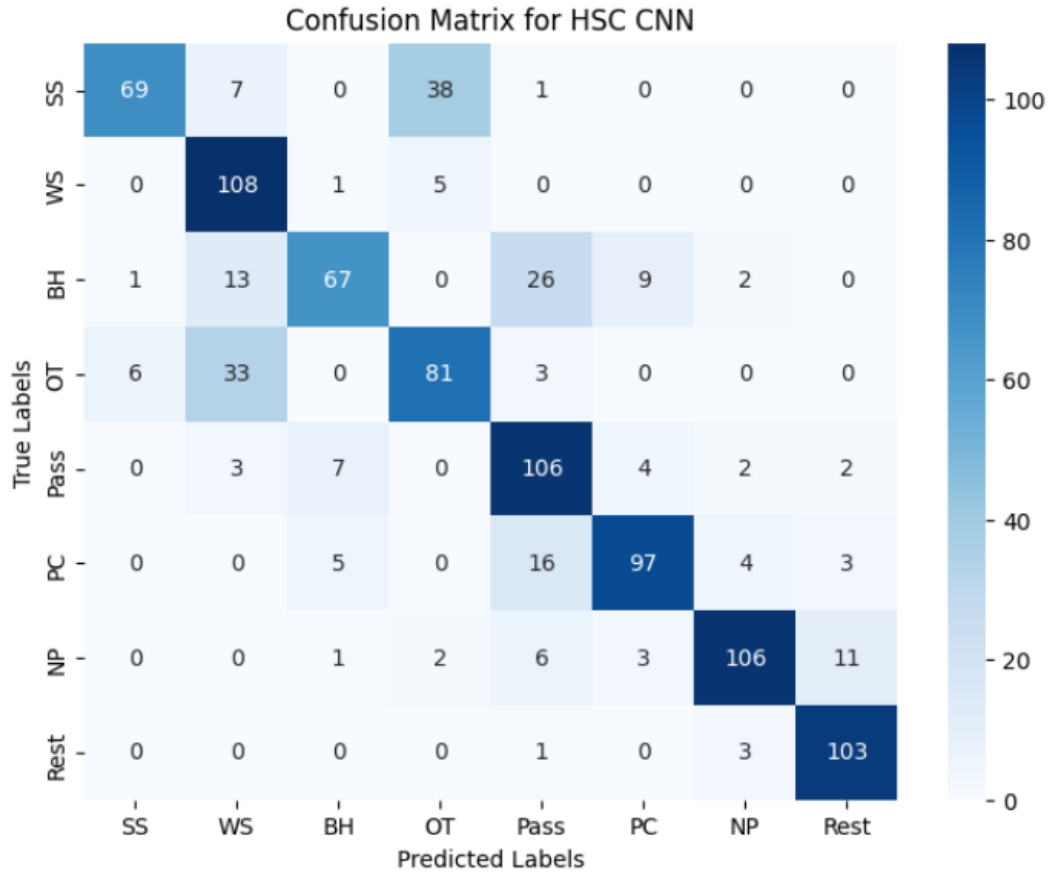
a)



b)


Confusion Matrix for SHSC CNN
True Labels
Predicted Labels
SS
WS
BH
OT
Pass
PC
NP
Rest
101 1 0 12 1 0 0 0
3 103 1 6 1 0 0 0
2 2 95 0 19 0 0 0
36 29 0 55 2 1 0 0
0 2 12 0 105 5 0 0
0 2 7 0 14 99 1 2
0 2 1 2 9 1 107 7
0 0 0 0 3 1 2 101
100
80
60
40
20
0

c)



Note. HSC = Hands Sensor Configuration; CNN = Convolutional Neural Network; SS = Slap Shot*;* WS = Wrist Shot; BH = Back Hand; OT = One-Timer*;* PC = Puck Control; NP = No Puck

5.3 Hybrid Model Performance

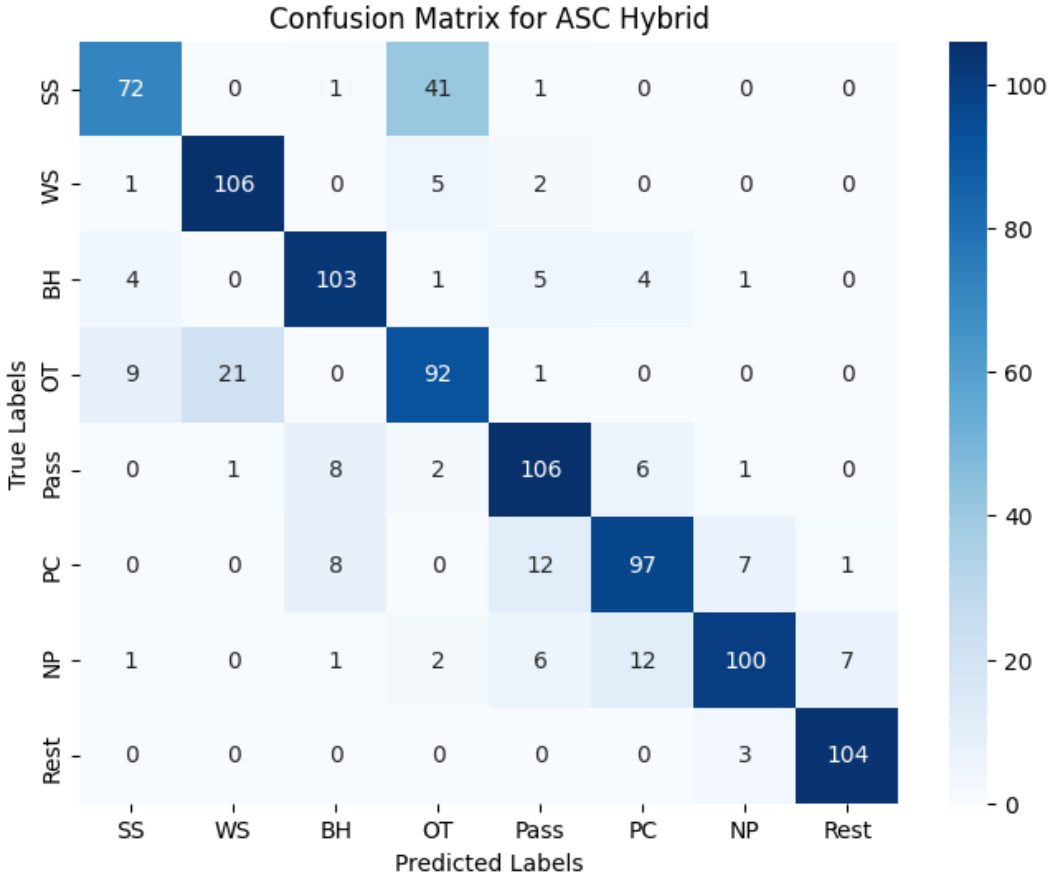
The Hybrid model displayed the best overall performance (Table 5). The ASC Hybrid model achieved an 81.6% F1 score, demonstrating the highest F1 score for the *rest* events (95%), and the lowest for the *one-timer* events (69.2%). The SHSC Hybrid model achieved an F1 score of 83.7%, demonstrating the highest F1 score for the *rest* events (95.4%) and the lowest for the *one-timer* events (69.9%). The HSC Hybrid model achieved an F1 score of 80.8%, demonstrating the highest F1 score for the *rest* events (94.1%) and the lowest for the *one-timer*

events (65%). Confusion matrices were generated for all three configurations, revealing the number of predicted labels vs the true labels (Figure 14).

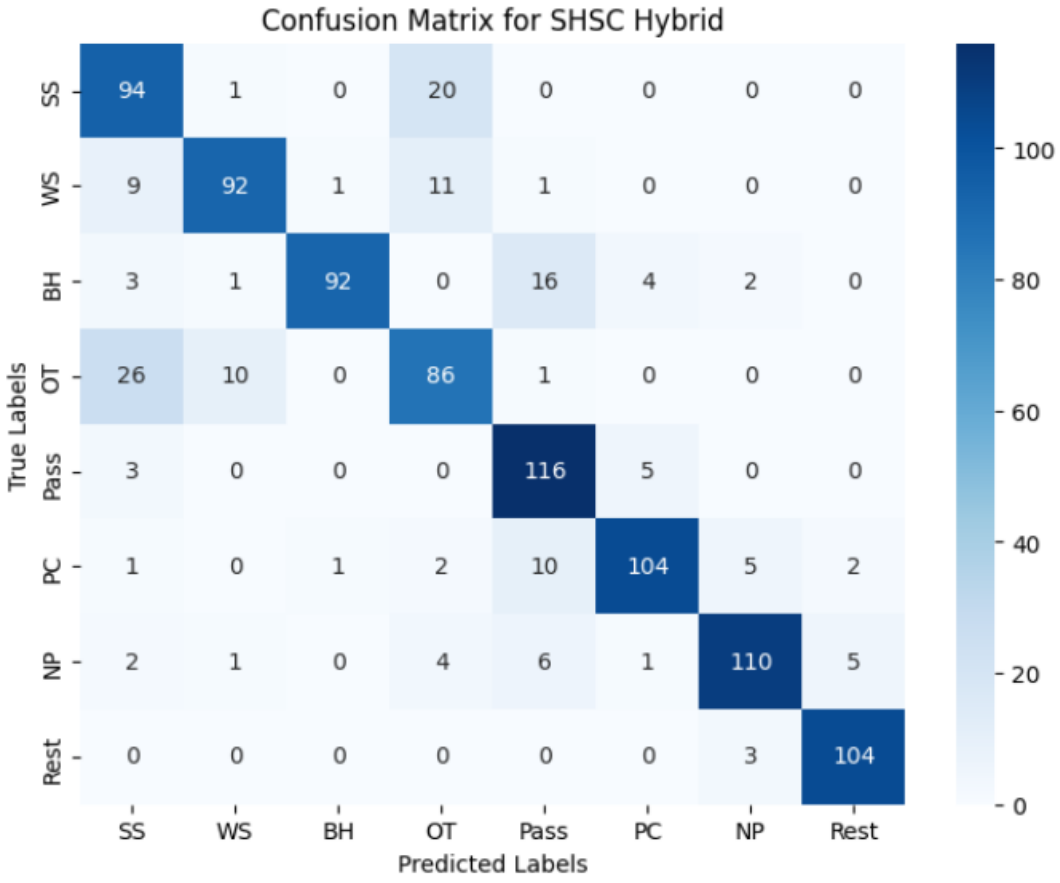
Figure 14

Confusion Matrices for the Hybrid Models with the ASC (a), the SHSC (b), and the HSC (c)

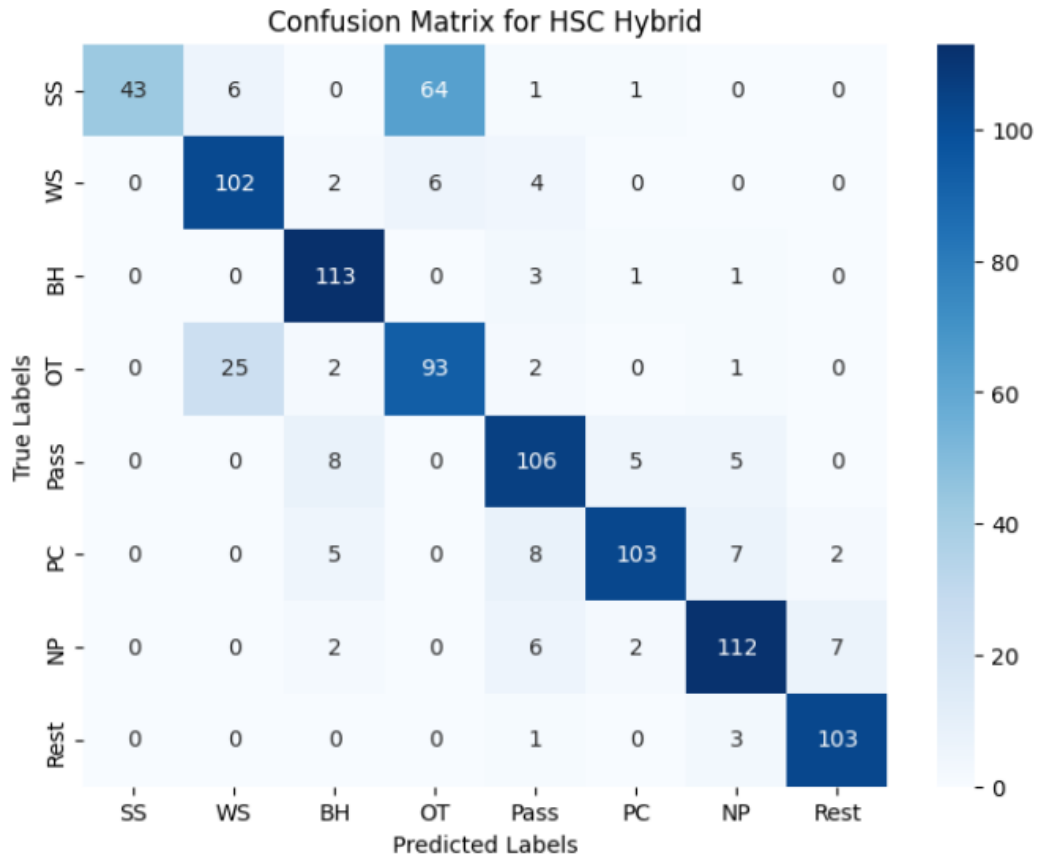
a)



b)


Confusion Matrix for SHSC Hybrid
True Labels
Predicted Labels
SS
WS
BH
OT
Pass
PC
NP
Rest
94 1 0 20 0 0 0 0
9 92 1 11 1 0 0 0
3 1 92 0 16 4 2 0
26 10 0 86 1 0 0 0
3 0 0 0 116 5 0 0
1 0 1 2 10 104 5 2
2 1 0 4 6 1 110 5
0 0 0 0 0 0 3 104
100
80
60
40
20
0

c)



Note. HSC = Hands Sensor Configuration; SS = Slap Shot; WS = Wrist Shot; BH = Back Hand; OT = One-Timer; PC = Puck Control; NP = No Puck

6. Discussion

This study sought to develop and compare the abilities of both traditional and novel deep learning approaches, along with three distinct IMU sensor configurations, in classifying eight ice hockey gameplay events. After training and testing three different deep learning model types with game-realistic IMU data, results revealed that the Hybrid model with the SHSC was the best-performing model setup, achieving an F1 score of 83.7%. The Hybrid models exhibited the highest mean F1 score across all three configurations (81.9%). The CNN models reached a slightly lower mean F1 score (79.5%), and the Transformer models, while still successful, attained the lowest mean F1 score (74.4%). The strong Hybrid model performance likely stems from the combined strengths of CNNs and Transformer models in processing complex time-series data. CNN's local feature extraction and positional encoding abilities, paired with the Transformers attention mechanism, helped the model to focus on the most relevant parts of the acceleration sequences and recognize fine-grained patterns within them. The novelty of these findings suggests that complex, highly dimensional accelerometer data, like that of ice hockey, requires robust deep learning models with detailed feature extraction and sequential recognition capabilities to accurately identify the various dynamic actions of the sport.

6.1 HAR in Ice Hockey

The results from this study are lower than Tremblay (2022), which used CNNs to classify similar ice hockey events from a 17-sensor configuration (F1 score = 97.1%) and a 2-hand sensor configuration (F1 score = 93.6%). One significant difference in Tremblay (2022) is that each shot was performed in a controlled setting with consistent movement patterns. The CNN models from that study were likely able to easily recognize the consistent patterns in the non-game

realistic acceleration signals. Furthermore, the CNN models developed in that study utilized three convolutional layers, compared to only two layers used in the current study. Increasing the number of convolutional layers allows the network to extract more abstract and hierarchical features, potentially enabling the model to better distinguish subtle differences between event types. Additionally, Tremblay (2022) tested only two sensor configurations, observing that the additional sensors in the configuration improved the CNN's classification score. Those observations remain consistent with the current study, which also found that additional sensors in the configuration led to a higher F1 score for the CNN models. While the Transformer models in this study also followed this trend, the Hybrid models did not and instead revealed the three-sensor configuration to be optimal. One potential explanation for this could be that the additional sensors added unnecessary data that "clutters" the model's ability to learn, and that the combination of the CNN and the Transformer was sufficient in extracting and highlighting relevant details from less data. Establishing a successful result with fewer sensors underscores the feasibility of conducting further ice hockey HAR studies beyond controlled settings.

While the current study proposed deep learning approaches for feature extraction and classification in ice hockey, other studies have explored alternative feature extraction methods and traditional machine learning techniques to automatically classify similar hockey events (e.g., *slap shots, wrist shots, hits, turns, time on ice, breaks*) from IMU data (Hardegger et al., 2015). Hardegger (2015) extracted numerous statistical (mean, min, and max stick rotational velocities) and frequency features (fast Fourier transform coefficients) from raw torso and hand IMU data from four professional hockey players. These features were used to train a Random Forest Classifier machine learning model, which achieved an overall F1 score of 70% (Hardegger et al., 2015). Using the same sensor configuration as in Hardegger (2015) (sternum and hands), the

deep learning approaches proposed in the current study achieved an overall F1 score of 83.7% (Table 6). Given that previous hockey HAR studies were limited by their reliance on highly controlled data, the current study took a novel approach and advanced the field by developing robust deep learning models trained on game-realistic accelerometer signals, outperforming earlier machine learning methods based on controlled data. Despite the lower classification scores than Tremblay (2022), this study demonstrated the advanced applicability of IMU-based event detection in potential real-world hockey environments.

6.2 HAR in Other Sports

Further studies have explored hybrid deep learning approaches and sensor configurations for IMU-based event classification in other sports, finding varying results. Khatun et al. (2022) proposed a CNN-LSTM hybrid model featuring an attention mechanism to classify 12 physical activities (e.g., *stand, sit, walk, cycle, jog, jump*, etc.) from smartphone IMU data, achieving a 98.76% accuracy (Khatun et al., 2022). The researchers suggested that the automated feature engine of the CNN and LSTMs could successfully extract temporal features from sequential accelerometer data. Using a similar hybrid model for soccer classification, a CNN-LSTM hybrid network was trained and tested using IMU data collected from the pelvis, thighs, and shins of eleven male soccer players who performed five soccer tasks in a controlled setting, achieving an accuracy of 98.25% (Cuperman et al., 2022). In contrast, basketball researchers employed wrist accelerometers to record various actions from 24 individuals in controlled drills, controlled warm-ups, and in-game settings (Hoelzemann et al., 2023). A CNN-LSTM model and an Attend-and-Discriminate (Transformer-like) model were trained on the controlled setting data and then evaluated on both the controlled setting data and the real game data. While both models achieved

F1 scores over 80% for locomotion activities (e.g., walking and running), they struggled on basketball-specific actions, scoring less than 45% for layups, passes, and rebounds. The model's performance dropped further when tested on the in-game data, with some scores below 20%. The researchers concluded that while the Attend-and-Discriminate (Transformer-like) model was slightly more robust, both models' performances significantly dropped due to the uncontrolled nature of the game, and that future work should focus on better generalizing these HAR models to the dynamic, uncontrolled sports environments (Hoelzemann et al., 2023). These findings align with the current study's aim of simulating an in-game hockey environment to better generalize future HAR models for in-game applications. The conclusions of Hoelzemann et al. (2023) reinforce the need for studies of this nature.

6.3 Event Classifications

A fundamental consideration in deep learning is that a model's ability is limited by the quality of the data that it is trained on. This limitation presented itself within the *one-timer* events in this study. Across all nine models, *one-timers* consistently had the lowest classification scores. They were often misclassified as either *slap* or *wrist shots* (Figure 12-14). This is likely a result of the inter- and intra-performance variability among participants. A *one-timer* is a shot in which a player performs a shot on their first contact with the puck; however, these shots can physically be executed with the same movement patterns as a *slap shot* or a *wrist shot*. This observation was maintained to be true in the data collection for this study. It could, therefore, be understood why the models frequently misclassified *one-timer* events as *slap shots* or *wrist shots.* It is plausible that a higher classification score would have been achieved if the shot labels were more representative of their actual motions (e.g., labelling a *one-timer* event as a *wrist shot* if it was

actually performed with a *wrist shot* motion). The models rarely misclassified *slap shots* as anything other than one-timers, indicating their ability to recognize the *slap shot* motion, which, as previously stated, is also characteristic of many *one-timers*. Future studies should be more specific about how events are defined and labelled based on their movement patterns, and should consider removing the *one-timer* label altogether. These changes would allow models to be trained on labelled data that is more representative of the actual movements displayed, thereby enabling the models to more efficiently and accurately learn the underlying patterns in the acceleration signals of the correct hockey actions, further increasing the models' accuracy.

Another common source of error was with the *passing* events (Figure 12-14). The *pass* execution also varied significantly across samples. Some *passes* were performed with minimal movement, some of the *passes* mimicked a stick-handling motion, some *passes* were larger movements that could have looked like a *wrist shot*, and others were performed off the participants' backhands. This variability in pass execution likely hindered the model's ability to detect distinct consistent patterns in the acceleration signals, resulting in frequent misclassification with other events that exhibited similar movement characteristics. It is worthwhile to note that passes in a live game greatly vary as well, further supporting the representativeness of this data compared to that of a real game.

All nine models achieved the highest classification score for *rest* events, despite there being fewer *rest* samples than any other event type. This was likely a result of the inherently stationary nature of the *rest* event, during which the player remained seated on the bench, in contrast to the dynamic, full-body movement characteristics of the other events. Hardegger (2015) also attempted to classify off-ice *rest* events with their machine learning model, achieving an F1 score of 86% for *rests*, compared to the 95.4% of the SHSC-Hybrid model in the current

study (Table 6). Similarly, in Hardegger (2015), the researchers stated that they attempted to classify whether a player was in possession of the puck or not, but they were unable to accomplish that task, so that method was discarded. In the current study, the SHSC-Hybrid model achieved an F1 score of 87% in classifying whether the player was in control of the puck (Table 6). These findings reveal both the strengths and challenges of the proposed deep learning models when applied to the collected dataset. They underscore the importance of consistent, movement-based event labelling to reduce ambiguity in hockey event data and enhance the performance of future hockey HAR models.

6.4 Transformer Learning

Recent work by Apple's Machine Learning Research division highlighted the highly sensitive nature of Transformer models (Zhai et al., 2023). Transformer networks initially distribute attention across many parts of the input, but as training progresses, attention entropy causes them to focus more narrowly on the most relevant tokens. The researchers found that attention entropy is strongly correlated to training stability and convergence. They observed that without careful tuning of the learning rate, attention entropy might fail to converge. A learning rate that is too small can result in unstable training and high variance in the loss, while a learning rate that is too large may lead to attention entropy collapse and eventual training divergence (Zhai et al., 2023). The learning rate is featured during backpropagation and controls how much the model's weights and biases update with respect to minimizing the loss score in response to the error after each forward pass (Chollet, 2018). To address these sensitivity issues within Transformers, researchers suggest fine-tuning hyperparameters such as the learning rate, warmup steps, initialization, and learning rate scheduler to improve training stability (Zhai et al., 2023).

From the hyperparameter tuning outputs in the current study (Figure 10), it was observed that the learning rate parameter had the most significant impact on the models' performances. High loss variance and convergence failures, as described in Zhai et al. (2023), were observed during the early model experimentation and learning rate tuning stage (Appendix D). A similar study using Transformers for HAR from IMU data featured finely tuned warmup steps and learning rate scheduling, attaining positive success in their model's classification score (99.2% accuracy) (Dirgová Luptáková et al., 2022). Warmup steps are the number of training steps it takes to increase from a small initial value to the target maximum learning rate (Dirgová Luptáková et al., 2022). This makes the model capable of handling higher target learning rates later in training, thereby enhancing overall stability and improving final model performance (Kalra & Barkeshli, 2024). The learning rate scheduler then gradually decreases the learning rate throughout the rest of the training, enabling the model to fine-tune its parameters more carefully and avoid large updates when it is close to convergence. Together, these learning rate mechanisms improve learning stability by reducing the model's sensitivity to sudden, large gradient updates both early and late in training, while also allowing them to find an optimal and generalizable solution.

6.4 Computational Demands

A key consideration in the development of these models is the time and computational resources required to run them. The Transformer networks in this study demanded significantly higher time and computational resources than the CNNs. The hyperparameter tuning, which involved tuning 8 hyperparameters for 20 trials of 20 epochs, took approximately 50-60 minutes to run, and training the final model with the best parameters took approximately 5 minutes to

complete. The total computational demands to run a Transformer model was roughly 38 GB of A100 GPU RAM. The convolutional layers in the hybrid models slightly alleviated the high time and computational burdens, requiring 38-48 minutes to tune twelve hyperparameters and 3-5 minutes for final model training, requiring roughly 15 GB of GPU RAM. The CNN models, on the other hand, took around 6 minutes to tune eight hyperparameters and roughly 30 seconds for model training, requiring only 1-3GB of GPU RAM. The reduced time and computational requirements could attract developers to choose CNNs for time series classification tasks, as it can be far less expensive to optimize a CNN model.

6.5 Practical Implications

Findings from this study will provide direction for the continued development of deep learning HAR sport models, as they hold significant potential for enhancing player and team development. Highlighting the strengths and weaknesses of the proposed models improves our understanding of effective model design for dynamic task classification and can guide future model development. With successful classification abilities using fewer sensors and game-simulated data, this study demonstrates the viability of deploying future models in real-game scenarios. Coaches and trainers would benefit from these advancements, as HAR models could facilitate the collection and analysis of detailed statistics from games and practices. With the current ability to accurately classify hockey-specific movements, future models may also have the ability to identify individual player movement patterns and detect elements that could be associated with injury risks or provide targeted biomechanical feedback to optimize performance. For example, future hockey HAR models could be developed to recognize a player's kinematic patterns before and after an injury to analyze performance deviations. With

these insights, a model could objectively suggest modifications to their joint sequencing to enhance shot performance or recommend kinematic adjustments to reduce the risk of skating-related injuries. Additionally, findings from this study could partner with the development of a live streamlining pipeline for real-time feedback, which could improve the efficiency and efficacy of training sessions or provide coaches with quicker and more detailed in-game insights. Overall, these algorithms have the potential to significantly enhance sports analytics, player health monitoring, and performance improvement strategies.

6.6 Limitations

While this study aimed to simulate a live game, data were still collected in a controlled environment with only two people on the ice at a time, as opposed to the twelve players in an actual game. The addition of opponents and teammates on the ice could alter the intensity of on-ice performance and the execution of the various tasks. Numerous deep learning techniques, tuning methods, and layer configurations were explored in this study. Although the final models demonstrated success, the experimentation was not exhaustive, and other preprocessing steps, model architectures, and hyperparameter settings may yield additional improvements. Additionally, the raw IMU data were preprocessed using Xsens' proprietary sensor fusion algorithm, featuring an extended Kalman filter. While this ensures consistent and reliable motion data, the closed nature of the algorithm restricts insight into how the original signals are transformed, potentially influencing downstream model performance.

6.7 Future Directions

Building on the findings of this study, several future directions to further the scope of hockey HAR research can be identified. Next steps include testing the proposed models on data collected in a live game, in which players are wearing all their equipment. Results from this would help identify areas within the models and data pipelines that require refinement to optimize accuracy and efficiency. Future studies should include a more specific event labelling process, specifically for *one-timers* and *passes,* to ensure consistent labels for each action. They should explore methods to increase the data set size used to train the model through various techniques, including additional data collection, data augmentation, focal loss, class weight adjustments, transfer learning, etc. Future studies should experiment with alternative data preprocessing techniques, like using a sliding window method for event segmentation, interpolating the data for consistent input sizes, and alternative data normalization techniques. Model design changes could also be explored, such as incorporating LSTM layers, incorporating learning rate mechanisms, expanding the hyperparameter search space, or designing a predictive model to predict an event before it occurs. Finally, exploring alternative sensor configurations could play a key role in preparing these models for deployment in live game environments by improving practicality and comfort. Future studies should investigate single-sensor setups that are seamlessly integrated with player gameplay, such as a chest-mounted sensor integrated into the shoulder pads, a smartwatch on a player's wrist, a sensor embedded into a player's glove, or stick-mounted sensors. Pursuing these approaches could significantly enhance the feasibility, scalability, and real-world adoption of wearable HAR systems in ice hockey.

7. Conclusion

This study explored the development of deep learning approaches for classifying eight ice hockey player events from game-realistic accelerometer data. Through a comprehensive model development and testing process, the Hybrid network combined with the SHSC was identified as the optimal approach, achieving an 83.7% F1 score. This study demonstrated the effectiveness and practicality of wearable sensors paired with neural networks to recognize detailed player movements in a game-simulated environment. The diverse participant population in the study, which featured male, female, elite, and recreational players, highlights the broad applicability of these models. Insights derived from this study can inform decisions on future deep learning-based approaches for identifying and analyzing player movement data. Future studies should explore the deployment of these HAR models in live-game scenarios to further advancements in hockey analytics, player health monitoring, and performance optimization.

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Appendix A

Ethics Certification

McGill University
Research Ethics Board Office
www.mcgill.ca/research/research/compliance/human

CERTIFICATE OF ETHICS APPROVAL

REB File Number:	23-04-009
Project Title:	Assessing the Efficacy of a Deep Learning Model in Classifying Ice Hockey Shots Using Game Realistic Data from Inertial Sensors
Faculty Principal Investigator:	Shawn Robbins
Department:	Physical & Occupational Therapy, School of
Sponsor/Funding Agency (if applicable):	NSERC and Bauer Ltd.

Research Team (if applicable):

Name	Affiliation
Philippe Dixon	University de Montreal
Andrew Phillips	McGill University
Shawn Robbins	McGill University
Philippe Renaud	McGill Univeristy

Approval Period:	
FROM	**TO**
03-Jul-2023	02-Jul-2024

The *REB-2* reviewed and approved this project by Full Board review in accordance with the requirements of the McGill University Policy on the Ethical Conduct of Research Involving Human Participants and the Tri-Council Policy Statement: Ethical Conduct For Research Involving Humans.

* Approval is granted only for the research and purposes described.
* The PI must inform the REB if there is a termination or interruption of their affiliation with the University. The McGill REB approval is no longer valid once the PI is no longer a student or employee.
* An **Amendment** form must be used to submit any proposed modifications to the approved research. Modifications to the approved research must be reviewed and approved by the REB before they can be implemented. Changes to funding or adding new funding to a previously unfunded study must be submitted as an Amendment.
* A **Continuing Review** form must be submitted before the above expiry date. Research cannot be conducted without a current ethics approval. Submit 2-3 weeks ahead of the expiry date.
A total of 5 renewals are permitted after which time a new application will need to be submitted.
* A **Termination** form must be submitted to inform the REB when a project has been completed or terminated.
* A **Reportable New Information** form must be submitted to report any unanticipated issues that may increase the risk level to participants or that may have other ethical implications or to report any protocol deviations that did not receive prior REB approval.
* The REB must be promptly notified of any new information that may affect the welfare or consent of participants.
* The REB must be notified of any suspension or cancellation imposed by a funding agency or regulatory body that is related to this study.
* The REB must be notified of any findings that may have ethical implications or may affect the decision of the REB.

Appendix B

Participant Consent Form

McGill
Department of Kinesiology and Physical Education

Information and Participant Consent Form

McGill Ice Hockey Research Group
Groupe de Recherche Hockey sur Glace

Researcher:
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Title of Project: Assessing the Efficacy of a Deep Learning Model in Classifying Ice Hockey Shots Using Game Realistic Data from Inertial Sensors

Sponsor(s): National Sciences and Engineering Research Council of Canada

Purpose of the study: The purpose of the study is to assess the ability of an AI deep learning algorithm in automatically classifying ice hockey events using acceleration data from inertial sensors worn in a game-realistic testing protocol. Developing a basis for AI based human activity recognition can provide players, trainers, teams, and companies with statistics and analysis methods not traditionally available to them.

This study also seeks to compare the model's accuracy when using data from all 17 sensors compared to using data from just the two hands sensors.

Deep learning is a way for computers to learn from information it receives and make decisions or predictions.

Inertial sensors provide a way to measure each body part's acceleration and joint angles during movements.

Study Procedures: The approximate time commitment for the study is between 1-1.5 hours in only one session (~20 minutes pre trial in locker room, ~ 40 minutes on ice, ~15 minutes post

trial in locker room). You will be asked to bring your own skates, gloves, and stick that you would typically use in a game, and a water bottle. You will be required not to wear a helmet, as there are sensors that will be placed directly on the head and a helmet would interfere however, given your high level of experience of hockey, the risk of falls is minimal.

For clothing, please bring shorts, a T-shirt or sleeveless shirt, track pants, and track jacket or sweater. All other equipment will be provided by the researcher. The data collection session will take place at the McGill McConnell area and will begin in locker room three. There you will be asked to fill this consent form, and a participant information form.

Once the forms are read, signed, and the study protocol is explained, several body measurements will be taken that will be inputted into the computer system for system calibration purposes (i.e., total height, knee height, arms span, etc.) . The 17 sensors will then be placed on you. 14 sensors will be taped directly to your skin with double sided tape and secured with medical tape. You will then put on a head band containing the head sensor. The Xsens tight fitted shirt will but put on next, and all the sensors will be attached and plugged into a small pack that fits in the back of the shirt. You can then put on your track pants, jacket, gloves, and skates, so that we can then tape the foot sensors to your skate laces. We will then escort you onto the ice surface.

When arriving on the ice, you will first go to the carpet to calibrate the sensors. For the calibration steps, you will stand still, walk about four meters, turn 180 degrees, and walk back to your original spot and stand still again until the calibration is complete. We will direct you through these steps during this calibration process. After calibration is complete, you will have five minutes to skate around the ice to warm up.

Testing protocol

The trial will begin with you sitting on the bench and a researcher will give you your first sequence to execute. For each sequence, you will be told three, one-word verbal commands informing you how to receive the puck (“pass”: receive a pass from the passer, or “pick”: pick up a puck that the passer leaves for you), what shot to take (“slap”, “wrist”, “back hand”, “one timer”) and what net to shoot at (“one”, “two”, “three”, “four”). After completing the specific shot at the given net, you will skate back towards the bench where a researcher will give you the commands for your next sequence. After completing five sequences in a row, you will come back on the bench and sit down for a two-minute rest break. After the two minutes of rest, you will get back on the ice for another block of five sequences. You will complete 50 sequences total. If you accidentally lose the puck during a sequence, take the wrong shot, or shoot at the wrong net, just continue the trial, as minor mistakes like that will not impact the study. When you have the puck, you are encouraged to stick handle around the obstacles, and attack the net as you would in a real game. You are also encouraged to pass at the self-passing aid at least once per block. You are encouraged do the tasks at a high effort while maintaining control, and to stick handle through the on-ice obstacles during each sequence. Before you begin the trial, the researcher will give a demonstration on so you can see an example of how to execute the testing protocol.

Voluntary Participation: Participation in research is voluntary, and it will not affect your position on any hockey team. You may withdraw at any point or refuse to participate in any part of the study for any reason. Any of your data that has been collected will be destroyed, unless you chose to permit for some of the data to be kept. Once the study has been published, it may be too late for you to withdraw your data. It can only be removed from future analysis and publications. The data will be de-identified after the completion of the data collection and analysis (i.e., 31/08/2024) after which the data can no longer be linked to you.

Potential Risks: You may experience minimal discomfort when removing the sensors with attached medical tape from your skin and may experience brief redness where the tape was placed, which will disappear within a few hours after the study. You may become fatigued throughout the testing session; however, you will be given adequate rest and water breaks and will not be asked to perform at an effort level higher than your normal level of physical exertion during a game. This risk is mitigated however by your high level of hockey experience and there only being one participant and one research on the ice per data collection session.

Potential Benefits: Benefits may not directly benefit you in the short term, however, results may provide insight into future hockey AI developments that can help to grow and support players, teams, analysts, equipment developers discover new biomechanical data collection and analysis methods not traditionally available.

Compensation: You will be compensated $40 in cash for your participation in this study. You will also be entered into a draw with other participants in this study for a chance to receive one of four custom MyBauer hockey sticks.

Confidentiality: All personal information (i.e., name, date of birth, height, weight, sex etc.) collected during the study will be kept on a password protected computer to maintain your confidentiality. Your personal information will not be disclosed in any manner in the dissemination of the results. During data collection, your data will be labelled using an identification code. De-identified data will be stored on the internal hard drive on the computer for further analysis and will be kept confidential by the research team (i.e., principal investigator, faculty supervisors and lab technician). With your consent, all de-identified data will be kept permanently in the Ice Hockey Biomechanics Laboratory data base by Dr. Shawn Robbins and published to a publicly available data base for use by other researchers for unspecified purposes. No identifying information will be uploaded on these databases except for age and sex. Your video images will not be uploaded to the public database.

Trials will be video recorded for visual analysis, solely for the use of the researcher. The footage will be kept on a password protected computer accessible only by the PI, supervisors, and co- investigators/lab technician.

With your consent, the researcher may take pictures with a camera while during the study. This may be included in the final thesis report for visual reference to demonstrate the tasks and methods used. With your consent, video recordings may also be used in the dissemination of the results. When used for this purpose, in both photo and video captures, your face, or any secondary identifiable information such as tattoos, piercings, birthmarks, or any identifiable

clothing (such as your name, team name and number on team apparel) will NOT be shown. To maintain confidentiality, both pictures and videos will be stored on the hard drive on the password protected computer of the lab.

Please check Yes or No to the following statements:

Yes: __ No: __ You consent to have pictures taken during testing and used in the dissemination of the results (no face or secondary identifiable information shown)

Yes: __ No: __ You consent for use of video recorded material by the principal investigator in dissemination of results (no face or secondary identifiable information shown)

Yes: __ No: __ You consent for your de-identified data to stored online (cloud data base) and used for future research and analysis purposes

If you have any questions, concerns, or complaints at any time, please contact the student researcher – (Andrew Phillips; andrew.phillips2@mail.mcgill.ca) of the faculty supervisor (Dr. Shawn Robbins; shawn.robbins@mail.mcgill.ca).

If you have any ethical concerns or complaints about your participation in this study, and want to speak with someone not on the research team, please contact the Associate Director, research Ethics at 514-398-6831 or lynda.mcneil@mcgill.ca citing REB file number 23-04-009.

Please sign below once you have read the above document and if you consent to participate in this research project. A copy of this consent form will be kept by the researcher.

Please sign below if you have read the above information and consent to participate in this study. Agreeing to participate in this study does not waive any of your rights or release the researchers from their responsibilities. To ensure the study is being conducted properly, authorized individuals, such as a member of the Research Ethics Board, may have access to your information. A copy of this consent form will be given to you and the researcher will keep a copy.

Participant's Name: (please print) _____

Participant's Signature: _____

Date: _____

Appendix C

Participant Information Form

Participant Information Form

Participant ID#: _____

Date of Birth: _____

Sex: _____

Year of hockey experience: _____

Highest Level Played: _____

Current Team: _____

Position (F/D): _____

Shot Handedness: _____

Stick Brand and Model: _____

Injury History:

1. In the past year, have you suffered any injuries? Have they prevented you from playing hockey? Explain.

_____

_____

_____

2. In the past year, have you experienced any injury to your nervous system? (e.g. concussion, damage to a nerve, etc.) Has it prevented you from playing hockey? Explain

_____

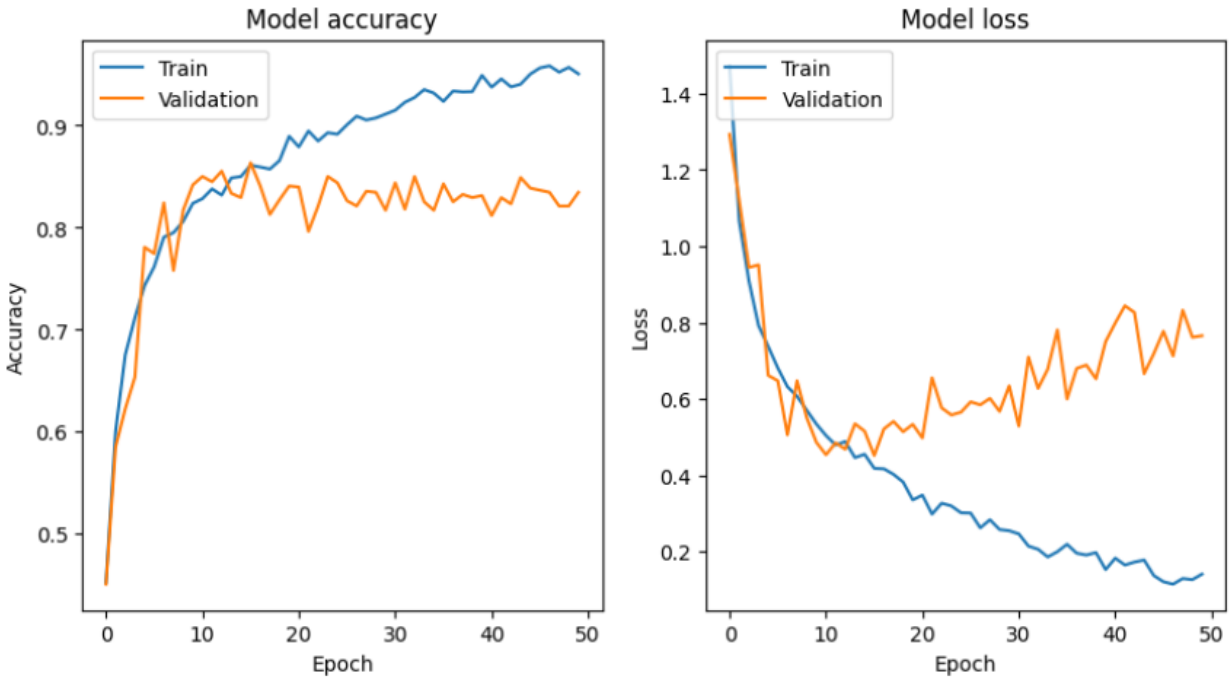
_____

_____

3. Is there any other reason why you believe you should not participate in this study? Explain.

Appendix D

Example of High Loss Variance and Model Divergence



Note. This plot shows the diverging loss curve (right) due to poor learning rate for the Hybrid HSC model during model experimentation and hyperparameter tuning. The final model corrected for this.